MATH 284 Notes

Linear Algebra

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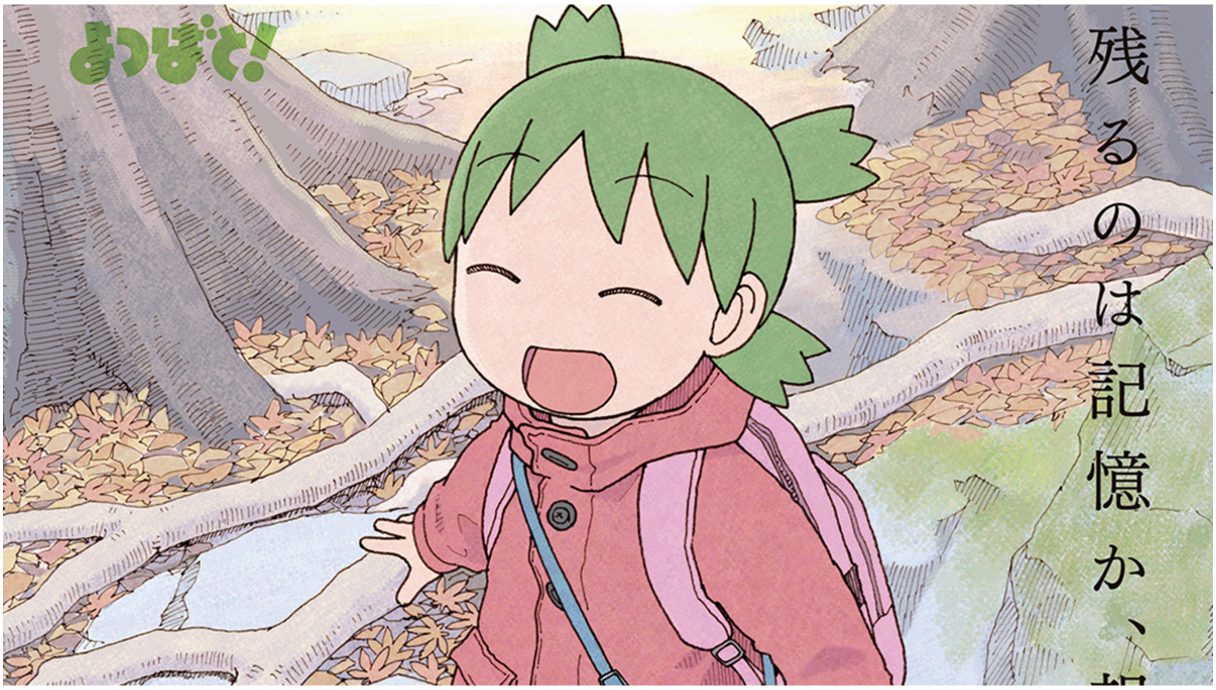

Yotsuba from Yotsuba&!

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1 Linear Equations in Linear Algebra

1.1 Systems of Linear Equations

- A **linear equation** with the variables $x_1, x_2, \ldots, x_n$ and coefficients $a_1, a_2, \ldots, a_n$ can be written as follows:

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

- A **system of linear equations** is a collection of numerous linear equations all using the same variables.

$$\begin{aligned} a_1x_1 + a_2x_2 + \cdots + a_nx_n &= k_1 \\ b_1x_1 + b_2x_2 + \cdots + b_nx_n &= k_2 \end{aligned}$$

- The solutions of a system of linear equations have three variants:
  - **No solution**. In $\mathbb{R}^2$, the linear equations would be parallel, non-overlapping lines.
  - **One solution**. In $\mathbb{R}^2$, the linear equations would be nonparallel lines.
  - **Infinitely many solutions**. In $\mathbb{R}^2$, the linear equations would be parallel, overlapping lines.

If a system of linear equations has a **solution set**, it is **consistent**. Otherwise, it is **inconsistent**.

Matrix Notation

We can display a system of linear equations with a matrix as follows.

$$\begin{aligned} 5x_1 + x_2 - 9x_3 &= 2 \\ x_1 \qquad + 4x_3 &= 7 \\ 6x_3 &= 1 \end{aligned} \qquad \begin{pmatrix} 5 & 1 & -9 \\ 1 & 0 & 4 \\ 0 & 0 & 6 \end{pmatrix}$$

- The matrix above is a **coefficient matrix**.
- Each column corresponding to a different variable.
- An **augmented matrix** has an augmented column of the system's constants, usually separated by a line for clarity.

$$\left(\begin{array}{ccc|c} 5 & 1 & -9 & 2 \\ 1 & 0 & 4 & 7 \\ 0 & 0 & 6 & 1 \end{array}\right)$$

Solving a Linear System

To solve a linear system, we can replace it with an equivalent system. We can create an equivalent system by applying an **elementary row operation** to the rows (or in equation form, equations).

Definition 1.1.1 (Elementary Row Operations)

1) **Replacement:** Replace one row by the sum of itself and a scalar multiple of another row.
2) **Interchange:** Swap two rows.
3) **Scaling:** Multiplies all entries of a row by a nonzero constant.

♣

- Thus, two matrices are **row equivalent** if there exists a sequence of elementary row operations that transforms one into the other.
- All elementary row operations are reversible.

1.2 Row Reduction and Echelon Forms

Definition 1.2.1 (Echelon Form)

A matrix is in **echelon form**, also known as **row echelon form**, if it has the following properties:
1) All nonzero rows are above any rows of all zeros.
2) Each leading (first) entry of a row is in a column to the right of the leading entry of the row above it.
3) All entries (in the same column) below a leading entry are zeros.

♣

Definition 1.2.2 (Reduced Row Echelon Form)

Furthermore, a matrix is in **reduced row echelon form (rref)** if it satisfies the previous three properties and the following properties:
1) The leading entry in each row is 1.
2) Each leading 1 is the only nonzero entry in its column.

♣

Theorem 1.2.3 (Uniqueness of the Reduced Echelon Form)

Each matrix is row equivalent to one reduced echelon form matrix.

♡

- Because the leading values of an echelon form matrix can be anything, they are not unique.

Pivot Positions

- A **pivot position** in a matrix is the location that corresponds to the location of a leading 1 in the matrix's reduced row echelon form.
- A **pivot column** is any column containing a pivot position.

The Row Reduction Algorithm

Algorithm 1.2.4 (Row Reduction Algorithm)

1) Begin with the leftmost nonzero column. This will be a pivot column with a pivot position at the top.
2) Select a nonzero entry in the pivot column as the pivot. If needed, swap rows to move it into the pivot position.
3) Use elementary row operations to turn all entries below the pivot into zeros.
4) Repeat steps (1-3) with each consecutive nonzero column with pivot positions one row below the previous one.
5) Starting with the rightmost pivot, using scaling, if necessary, to convert each pivot to a 1.

♠

- When performing an elementary row operation, we refer to each n^{th} row as R_n.

$$\begin{pmatrix} 0 & 8 \\ 3 & 5 \end{pmatrix} \begin{matrix} R_1 \\ R_2 \end{matrix} \xrightarrow[R_1 \leftrightarrow R_2]{} \begin{pmatrix} 3 & 5 \\ 0 & 8 \end{pmatrix} \xrightarrow[R_1 - \frac{5}{8} R_2 \to R_1]{} \begin{pmatrix} 3 & 0 \\ 0 & 8 \end{pmatrix} \xrightarrow[\frac{1}{3} R_1 \to R_1]{} \begin{pmatrix} 1 & 0 \\ 0 & 8 \end{pmatrix} \xrightarrow[\frac{1}{8} R_2 \to R_1]{} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Solutions of Linear Systems

Applying the row reduction algorithm to an augmented matrix leads directly to a solution set for the associated linear system.

- **Basic variables** are variables whose coefficients are in a pivot column.
- **Free variables** are variables whose coefficients aren't in a pivot column.

If a system is consistent and has only one solution, then there will be no free variables.

If a system is consistent and has infinitely many solutions, then a solution set can be found by *finding the basic variables in terms of the free variables.*

If a system is inconsistent, then the solution set is empty.

Existence and Uniqueness

Theorem 1.2.5 (Existence and Uniquness Theorem)

A linear system is consistent ⇔ the corresponding augmented matrix's rightmost column is *not* a pivot column.

If a linear system is consistent, then the solution set contains either (i) a unique solution, when there are no free variables, or (ii) infinitely many solutions, when there is at least one free variable.

1.3 Vector Equations

A matrix with one column is known as a **column vector**, or just a **vector**.

- Vectors are generally denoted using boldface lowercase letters:

$$\mathbf{u} = \begin{pmatrix} 5 \\ 2 \\ 6 \end{pmatrix} \qquad \mathbf{v} = \begin{pmatrix} 19 \\ -5 \end{pmatrix} \qquad \mathbf{w} = \begin{pmatrix} -2 \\ 0 \\ 23 \\ -9 \end{pmatrix}$$

- Sometimes, vectors are used to express columns of matrix.

In general, the set of all vectors across some n-dimensional coordinate space is $\mathbb{R}^n$. We will often work with vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$ when geometric interpretations are needed, but going beyond the third dimension is not uncommon.

Geometric Vector Descriptions

For vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$, we can interpret vectors $\mathbf{u} = \begin{pmatrix} a \\ b \end{pmatrix}$ and $\mathbf{v} = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$ as directed line segments from the origin to the points (a, b) and (a, b, c), respectively.

Vectors in $\mathbb{R}^n$

Property 1.3.1 (Algebraic Properties of $\mathbb{R}^n$)

$\forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in \mathbb{R}^n$ and all scalars c and d:

1) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
2) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
3) $\mathbf{u} + \mathbf{0} = \mathbf{0} + \mathbf{u} = \mathbf{u}$
4) $\mathbf{u} + (-\mathbf{u}) = -(\mathbf{u}) + \mathbf{u} = \mathbf{0}$
5) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
6) $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
7) $c(d\mathbf{u}) = (cd)\mathbf{u}$
8) $1\mathbf{u} = \mathbf{u}$

Linear Combinations

A **linear combination** of $\mathbb{R}^n$ vectors $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$ is defined as follows:

$$\mathbf{y} = c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p$$

where scalars $c_1, c_2, \ldots, c_p$ are the **weights**.

- We can express infinite sets of vectors as linear combinations of finite sets of vectors.
- Infinitely large solution sets are expressed using linear combinations.

Definition 1.3.2 (Span)

If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$ are vectors in $\mathbb{R}^n$, then the set of all linear combinations of those vectors is denoted by $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ and is called the **subset of $\mathbb{R}^n$ spanned by** $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$.

- The question of whether or not some vector $\mathbf{b}$ is in $\text{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is equivalent to whether or not the following vector equation has a solution:

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_p\mathbf{v}_p = \mathbf{b}$$

- Relating this back to chapter 1.2, this is equivalent to this linear system:

$$\begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_p \end{pmatrix}$$

Geometric Span Descriptions

Let $\mathbf{u}$ and $\mathbf{v}$ be some nonzero vectors in $\mathbb{R}^3$.
- $\text{Span}\{\mathbf{u}\}$ is a line straight through $\mathbf{0}$ and $\mathbf{u}$.
- $\text{Span}\{\mathbf{u}, \mathbf{v}\}$ is a plane that $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{0}$ lie on.

These interpretations stem from the spans containing *all linear combinations of the vectors* in these subsets of $\mathbb{R}^3$.

1.4 The Matrix Equation $A\mathbf{x} = \mathbf{b}$

If A is an $m \times n$ matrix with columns $\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n$, and if $\mathbf{x}$ is a vector in $\mathbb{R}^n$, then the product $A\mathbf{x}$ is some vector $\mathbf{b} \in \mathbb{R}^m$ given by

$$A\mathbf{x} = \begin{pmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n = \mathbf{b}$$

- The form $A\mathbf{x} = \mathbf{v}$ is known as the **matrix equation**, as they can be analogous to linear systems.
- Matrices are often denoted by uppercase letters and are sometimes bolded.

Existence of Solutions

Theorem 1.4.1

Let A be an $m \times n$ matrix. Then, the following statements are equivalent.

a) $\forall \mathbf{b} \in \mathbb{R}^m$, the equation $A\mathbf{x} = \mathbf{b}$ has a solution.
b) Each $\mathbf{b} \in \mathbb{R}^m$ is a linear combination of the columns of A.
c) The columns of A span $\mathbb{R}^m$.
d) A has a pivot position in every row.

Caution

The matrix A of the matrix equation is a coefficient matrix, not an augmented matrix.

Properties of the Matrix-Vector Product

Theorem 1.4.2

If A is an $m \times n$ matrix, $\mathbf{u}$ and $\mathbf{v}$ are vectors in $\mathbb{R}^n$, and c is a scalar, then:

a) $A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$
b) $A(c\mathbf{u}) = c(A\mathbf{u})$

1.5 Solution Sets of Linear Systems

Homogenous Linear Systems

A linear system is **homogenous** if it can be written in the form $A\mathbf{x} = \mathbf{0}$ given some $m \times n$ matrix A and the $\mathbb{R}^m$ zero vector $\mathbf{0}$.

- A homogenous system will always have at least one solution, specifically the **trivial solution**, $\mathbf{x} = \mathbf{0}$.
- Generally, we are interested in whether or not a homogenous system has a **nontrivial solution**.

Proposition 1.5.1

The homogenous equation $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution $\Leftrightarrow$ the equation has at least one free variable.

Example 1.5.2

Solve the homogenous linear system below:

$$\begin{aligned} 8x_1 + 4x_2 + 5x_3 &= 0 \\ 2x_1 + 4x_2 \phantom{{}+5x_3} &= 0 \\ 3x_1 + 2x_2 + x_3 &= 0 \end{aligned}$$

$$\left(\begin{array}{ccc|c} 8 & 4 & 5 & 0 \\ 2 & 4 & 0 & 0 \\ 3 & 2 & 1 & 0 \end{array}\right) \xrightarrow[R_2-\frac{1}{4}R_1\to R_2]{} \left(\begin{array}{ccc|c} 8 & 4 & 5 & 0 \\ 0 & 3 & -\frac{5}{4} & 0 \\ 3 & 2 & 1 & 0 \end{array}\right) \xrightarrow[R_3-\frac{3}{8}R_1\to R_3]{} \left(\begin{array}{ccc|c} 8 & 4 & 5 & 0 \\ 0 & 3 & -\frac{5}{4} & 0 \\ 0 & -\frac{1}{2} & -\frac{7}{8} & 0 \end{array}\right)$$

$$\xrightarrow[R_1-\frac{4}{3}R_2\to R_1]{} \left(\begin{array}{ccc|c} 8 & 0 & \frac{20}{3} & 0 \\ 0 & 3 & -\frac{5}{4} & 0 \\ 0 & -\frac{1}{2} & -\frac{7}{8} & 0 \end{array}\right) \xrightarrow[R_3+\frac{1}{6}R_2\to R_3]{} \left(\begin{array}{ccc|c} 8 & 0 & \frac{20}{3} & 0 \\ 0 & 3 & -\frac{5}{4} & 0 \\ 0 & 0 & -\frac{13}{12} & 0 \end{array}\right)$$

$$\xrightarrow[-\frac{12}{13}R_3\to R_3]{} \left(\begin{array}{ccc|c} 8 & 0 & \frac{20}{3} & 0 \\ 0 & 3 & -\frac{5}{4} & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) \xrightarrow[R_1-\frac{20}{3}R_3\to R_1]{} \left(\begin{array}{ccc|c} 8 & 0 & 0 & 0 \\ 0 & 3 & -\frac{5}{4} & 0 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$$\xrightarrow[R_2+\frac{5}{4}R_3\to R_1]{} \left(\begin{array}{ccc|c} 8 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) \xrightarrow[\frac{1}{8}R_1\to R_1]{} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) \xrightarrow[\frac{1}{3}R_2\to R_2]{} \left(\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

Because there exists no free variables, the only solution is the trivial solution, $\mathbf{x} = \mathbf{0}$.

Parametric Vector Form

An *explicit description* of a solution set is often written in parametric vector form:

$$\mathbf{x} = s\mathbf{u} + t\mathbf{v}$$

where s and t are some real numbers.

Example 1.5.3

Describe the solutions of $A\mathbf{x} = \mathbf{0}$ in parametric vector form given $A = \begin{pmatrix} 1 & 2 & -2 & 5 \\ 0 & 1 & -5 & 3 \end{pmatrix}$.

$$\left(\begin{array}{cccc|c} 1 & 2 & -2 & 5 & 0 \\ 0 & 1 & -5 & 3 & 0 \end{array}\right) \xrightarrow[R_1 - 2R_2 \to R_1]{} \left(\begin{array}{cccc|c} 1 & 0 & 8 & -1 & 0 \\ 0 & 1 & -5 & 3 & 0 \end{array}\right)$$

$$\therefore x_1 + 8x_3 - x_4 = 0$$

$$x_2 - 5x_3 + 3x_3 = 0$$

$$\mathbf{x} = x_3 \begin{pmatrix} -8 \\ 5 \\ 1 \\ 0 \end{pmatrix} + x_4 \begin{pmatrix} 1 \\ -3 \\ 0 \\ 1 \end{pmatrix}$$

◇

Solutions of Nonhomogeneous Systems

When a nonhomogeneous linear system has infinitely many solutions, the general solution can be expressed in parametric vector form as *one solution to the system plus the general solution to the analogous homogeneous linear system.*

- Geometrically, solution sets between corresponding homogeneous and nonhomogeneous linear systems are *parallel*. Essentially, the solution set for any $A\mathbf{x} = \mathbf{b}$ is translated by $\mathbf{b}$ from the solution set for $A\mathbf{x} = \mathbf{0}$.

1.6 Applications of Linear Systems

1.7 Linear Independence

Definition 1.7.1

An indexed set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ in $\mathbb{R}^n$ is **linearly independent** if the vector equation

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \cdots + c_p\mathbf{v}_p = 0$$

only has the trivial solution. Otherwise, the set is **linearly dependent**.

♣

Linear Independence of Matrix Columns

Notice how the definition for the linear dependence is analogous to the matrix equation $A\mathbf{x} = \mathbf{0}$. Recognizing each column of A as a vector, we can say that if $A\mathbf{x} = \mathbf{0}$ only has the trivial solution, *then its columns are linearly independent.*

- If we want to test the linear dependence of a set of vectors, we can let them be the columns of a matrix, then solve the corresponding homogeneous system.
- A set of one vector is linearly independent ⇔ the vector is not the zero vector.

Sets of Two or More Vectors

Theorem 1.7.2 (Characterization of Linearly Dependent Sets)

An indexed set $S = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ of at least two vectors is linearly dependent ⇔ at least one of the vectors in S is a linear combination of the others.

If S is linearly dependent and $\mathbf{v}_1 \neq \mathbf{0}$, then there exists some $\mathbf{v}_j\ \forall j > 1$ such that $\mathbf{v}_j$ is a linear combination of the previous vectors.

♡

1.8 Linear Transformations

A **transformation**, **function**, or **mapping** T from $\mathbb{R}^n$ to $\mathbb{R}^m$ is a rule that maps each vector $\mathbf{x} \in \mathbb{R}^n$ to a vector $T(\mathbf{x}) \in \mathbb{R}^m$.

- This mapping is generally denoted $T : \mathbb{R}^n \to \mathbb{R}^m$.
- $\mathbb{R}^n$ is the **domain**.
- $\mathbb{R}^m$ is the **codomain**.
- $T(\mathbf{x})$ is the **image** of $\mathbf{x}$ under T.
- The set of all images $T(\mathbf{x})$ is the **range**.

Matrix Transformations

Many transformations are associated with matrix multiplication. A **matrix transformation** is generally denoted $\mathbf{x} \mapsto A\mathbf{x}$ where A is the **standard matrix** of the transformation. If this matrix transformation is $T : \mathbb{R}^n \to \mathbb{R}^m$, then $T(\mathbf{x}) = A\mathbf{x}$.

- In this case, because the domain is $\mathbb{R}^n$ and the codomain is $\mathbb{R}^m$, the standard matrix A must be $m \times n$.

Linear Transformations

Definition 1.8.1 (Linear Transformation)

A transformation T is **linear** if

i) $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ $\forall \mathbf{u}, \mathbf{v}$ in the domain of T.

ii) $T(c\mathbf{u}) = cT(\mathbf{u})$ $\forall$ scalars c and $\mathbf{u}$ in the domain of T.

♣

- Essentially, a linear transformation requires a standard matrix A, as these properties stem from the properties of mapping $\mathbf{x} \mapsto A\mathbf{x}$:

$$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v}$$
$$A(c\mathbf{u}) = cA\mathbf{u}$$

As a result, we can also ascertain two more properties of linear transformations.

- Given a linear transformation T, scalars c and d, and vectors $\mathbf{u}$ and $\mathbf{v}$ in the T's domain:

$$T(\mathbf{0}) = \mathbf{0}$$
$$T(c\mathbf{u} + d\mathbf{v}) = cT(\mathbf{u}) + dT(\mathbf{v})$$

1.9 The Matrix of a Linear Transformation

Earlier, we asserted that every linear transformation requires some standard matrix A. We can prove this using identity matrix and **standard basis vectors** $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$ (the columns of I, essentially).

Theorem 1.9.1

Let $T : \mathbb{R}^n \to \mathbb{R}^m$ be a linear transformation. Then, there exists a unique matrix A such that

$$T(\mathbf{x}) = A\mathbf{x} \quad \forall \mathbf{x} \in \mathbb{R}^n$$

In fact, A is an $m \times n$ matrix whose j^{th} column is $T(\mathbf{e}_j)$, where $\mathbf{e}_j$ is the j^{th} column of I_n.

$$A = \begin{pmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{pmatrix}$$

Proof.

$$\begin{aligned} T(\mathbf{x}) &= T(x_1\mathbf{e}_1 + x_2\mathbf{e}_2 + \cdots + x_n\mathbf{e}_n) \\ T(\mathbf{x}) &= x_1 T(\mathbf{e}_1) + x_2 T(\mathbf{e}_2) + \cdots + x_n T(\mathbf{e}_n) \\ T(\mathbf{x}) &= \begin{pmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \\ T(\mathbf{x}) &= A\mathbf{x} \end{aligned}$$

□

- Essentially, we can define any standard matrix for a linear transformation based on what they do to standard basis vectors.

Geometric Linear Transformations of $\mathbb{R}^2$

For each of the following transformations, let A denote the standard matrix. Additionally, imagine that basis vectors $\mathbf{e}_1$ and $\mathbf{e}_2$ form a *unit square.*

- The **identity transformation** maps a vector to itself. Thus, we can base all other geometric transformations on this transformation.

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

- **Reflection transformations** reflect a vector over an axis.

Over the x-axis

$$A = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Over the y-axis

$$A = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$$

- **Contraction** and **dilation transformations** scale a vector by some constant k.

Horizontal contraction if $k < 1$
Horizontal expansion if $k > 1$

$$A = \begin{pmatrix} k & 0 \\ 0 & 1 \end{pmatrix}$$

Vertical contraction if $k < 1$
Vertical expansion if $k > 1$

$$A = \begin{pmatrix} 1 & 0 \\ 0 & k \end{pmatrix}$$

- **Shear transformations** slant a vector up to some constant k.

Horizontal shear for $k \neq 0$

$$A = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$$

Vertical shear for $k \neq 0$

$$A = \begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}$$

- **Projection transformations** project a vector onto an axis.

Onto the x-axis

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

Onto the y-axis

$$A = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

- **Rotational transformations** rotate vectors by some angle θ.

$$A = \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix}$$

> **Note**
>
> When performing a series of geometric transformations, left multiply for each consecutive transformation.

One-to-One and Onto

Definition 1.9.2

A mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ is **onto** or **surjective** if each $\mathbf{b} \in \mathbb{R}^m$ is the image of *at least one* $\mathbf{x} \in \mathbb{R}^n$.

♣

- Equivalently, T is onto if its codomain and range are equal.

Definition 1.9.3

A mapping $T : \mathbb{R}^n \to \mathbb{R}^m$ is **one-to-one** or **injective** if each $\mathbf{b} \in \mathbb{R}^m$ is the image of *at most one* $\mathbf{x} \in \mathbb{R}^n$.

♣

- Equivalently T is one-to-one if $T(\mathbf{x}) = \mathbf{b}$ has either a unique solution or no solution.

When a mapping is both onto *and* one-to-one, it is **bijective**.

2 Matrix Algebra

2.1 Matrix Operations

Scalar multiples of matrices and matrix addition is done by component. Thus, matrices may only be added together if they have the same size.

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \qquad B = \begin{pmatrix} b_{11} & \cdots & b_{1n} \\ \vdots & \ddots & \vdots \\ b_{m1} & \cdots & b_{mn} \end{pmatrix}$$

$$A + B = \begin{pmatrix} a_{11} + b_{11} & \cdots & a_{1n} + b_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} + b_{m1} & \cdots & a_{mn} + b_{mn} \end{pmatrix}$$

Theorem 2.1.1

Let A, B, and C be $m \times n$ matrices, and let r and s be scalars.

a) $A + B = B + A$
b) $(A + B) + C = A + (B + C)$
c) $A + 0 = A$
d) $r(A + B) = rA + rB$
e) $(r + s)A = rA + sA$
f) $r(sA) = (rs)A$

Matrix multiplication is defined similarly to matrix-vector multiplication. This hinges on the fact that $(AB)\mathbf{x} = A(B\mathbf{x})$ given that A is $m \times n$, B is $n \times p$, and $\mathbf{x} \in \mathbb{R}^p$.

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} = \begin{pmatrix} \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \end{pmatrix} \qquad B = \begin{pmatrix} b_{11} & \cdots & b_{1p} \\ \vdots & \ddots & \vdots \\ b_{n1} & \cdots & b_{np} \end{pmatrix} = \begin{pmatrix} \mathbf{b}_1 & \mathbf{b}_2 & \cdots & \mathbf{b}_p \end{pmatrix}$$

$$AB = \begin{pmatrix} A\mathbf{b}_1 & A\mathbf{b}_2 & \cdots & A\mathbf{b}_p \end{pmatrix}$$

- In this case, AB is a valid product because the number of columns in A is equal to the number of rows in B. The resulting matrix is $m \times p$.

Caution

Matrix multiplication is not necessarily commutative.

Properties of Matrix Multiplication

Theorem 2.1.2

Let A be an $m \times n$ matrix, and let B and C have valid sizes for the indicated sums and products.

a) $A(BC) = (AB)C$
b) $A(B + C) = AB + AC$
c) $(B + C)A = BA + CA$
d) $r(AB) = (rA)B = A(rB)$ for any scalar r
e) $I_m A = A I_n = A$

The Transpose of a Matrix

Given an $m \times n$ matrix A, the **transpose** of A, denoted A^{T}, is an $n \times m$ matrix whose columns are formed from the corresponding rows in A.

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \qquad A^{\mathsf{T}} = \begin{pmatrix} a_{11} & \cdots & a_{m1} \\ \vdots & \ddots & \vdots \\ a_{1n} & \cdots & a_{mn} \end{pmatrix}$$

Theorem 2.1.3

Let A and B be matrices with valid sizes for the following sums and products.

a) $\left(A^{\mathsf{T}}\right)^{\mathsf{T}} = A$
b) $(A + B)^{\mathsf{T}} = A^{\mathsf{T}} + B^{\mathsf{T}}$
c) $(rA)^{\mathsf{T}} = rA^{\mathsf{T}}$ for any scalar r
d) $(AB)^{\mathsf{T}} = B^{\mathsf{T}} A^{\mathsf{T}}$

2.2 The Inverse of a Matrix

An $n \times n$ **(square)** matrix A is **invertible** or **nonsingular** if there exists an $n \times n$ matrix C such that

$$CA = AC = I$$

where C is the **inverse** of A. Thus, C may also be denoted as A^{-1}.

- Due to the nature of matrix multiplication, rectangular matrices can be **left invertible** or **right invertible**, but *not both*.

Theorem 2.2.1

Let $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$. If $ad - bc \neq 0$, then A is invertible and

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

- The quantity $ad - bc$ is the **determinant** of A, denoted $\det A$. Determinants of larger matrices will be covered in chapter 3.

Theorem 2.2.2

If A is an invertible $n \times n$ matrix, then $\forall \mathbf{b} \in \mathbb{R}^n$, the equation $A\mathbf{x} = \mathbf{b}$ has a solution.

Proof. Suppose A is any invertible $n \times n$ matrix and then $\mathbf{b}$ is some vector in $\mathbb{R}^n$.

$$\begin{aligned} A\mathbf{x} &= \mathbf{b} \\ A^{-1}A\mathbf{x} &= A^{-1}\mathbf{b} \\ I\mathbf{x} &= A^{-1}\mathbf{b} \\ \mathbf{x} &= A^{-1}\mathbf{b} \end{aligned}$$

□

Theorem 2.2.3

Let A and B be invertible matrices.

a) $\left(A^{-1}\right)^{-1} = A$

b) $(AB)^{-1} = B^{-1}A^{-1}$

c) $\left(A^{\mathsf{T}}\right)^{-1} = \left(A^{-1}\right)^{\mathsf{T}}$

Proof. Suppose that A and B are invertible matrices.

$$\begin{aligned} A^{-1}\left(A^{-1}\right)^{-1} &= I \\ \left(A^{-1}\right)^{-1} &= AI \\ \left(A^{-1}\right)^{-1} &= A \end{aligned}$$

$$\begin{aligned} AB(AB)^{-1} &= I \\ B(AB)^{-1} &= A^{-1}I \\ (AB)^{-1} &= B^{-1}A^{-1}I \\ (AB)^{-1} &= B^{-1}A^{-1} \end{aligned}$$

$$\begin{aligned} \left(A^{-1}\right)^{\top}A^{\top} &= \left(AA^{-1}\right)^{\top} \\ \left(A^{-1}\right)^{\top}A^{\top} &= I^{\top} \\ \left(A^{-1}\right)^{\top}A^{\top} &= I \end{aligned} \qquad \begin{aligned} A^{\top}\left(A^{-1}\right)^{\top} &= \left(A^{-1}A\right)^{\top} \\ A^{\top}\left(A^{-1}\right)^{\top} &= I^{\top} \\ A^{\top}\left(A^{-1}\right)^{\top} &= I \end{aligned}$$

□

Elementary Matrices

An **elementary matrix** is a matrix obtained from performing a single elementary row operation on an identity matrix, generally denoted E.

$$E_1 = \begin{pmatrix} 1 & 0 & -5 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad E_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \qquad E_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 9 \end{pmatrix}$$

$$R_1 - 5R_3 \to R_1 \qquad R_2 \leftrightarrow R_3 \qquad 9R_3 \to R_3$$

- They are particularly useful in proofs for representing elementary row operations, as left multiplying by an elementary matrix corresponds to one elementary row operation.
- Because elementary row operations are reversible, elementary matrices are invertible.

Theorem 2.2.4

An $n \times n$ matrix A is invertible $\Leftrightarrow$ A is row equivalent to I_n. In other words, there exists a sequence of elementary row operations that reduces A to I_n. Additionally, that sequence also reduces I_n to A^{-1}.

Proof. Suppose A is an invertible $n \times n$ matrix. Then, $\forall b \in \mathbb{R}^n$, there exists a unique solution to $\mathbf{x} = A^{-1}\mathbf{b}$ to $A\mathbf{x} = \mathbf{b}$. Because A is square, the main diagonal represents its pivot positions, implying that A can be reduced to I_n.

Now, suppose that A is row equivalent to I_n. Thus, there exists a sequence of elementary row operations represented by elementary matrices $E_1, E_2, \ldots, E_p$ that reduce A to I_n. Because elementary matrices are invertible, $E_p \cdots E_2 E_1$ is invertible. By definition, $E_p \cdots E_2 E_1$ is an inverse for A. Thus, A is invertible. Additionally:

$$\begin{aligned} E_p \cdots E_2 E_1 A &= I_n \\ E_p \cdots E_2 E_1 A A^{-1} &= I_n A^{-1} \\ E_p \cdots E_2 E_1 I_n &= A^{-1} \end{aligned}$$

□

An Algorithm for A^{-1}

We can augment any $n \times n$ matrix A with I, then reduce A, if possible, to I. By the previous theorem, it follows that the augmented columns will form A^{-1} once A is reduced to I.

$$(A \;\; I) \sim \left(I \;\; A^{-1}\right)$$

Example 2.2.5

Find the inverse of $A = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 4 & -3 & 8 \end{pmatrix}$, if it exists.

$$\left(\begin{array}{ccc|ccc} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_1 \leftrightarrow R_2]{} \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 4 & -3 & 8 & 0 & 0 & 1 \end{array}\right) \xrightarrow[R_3 - 4R_1 \to R_3]{} \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & -3 & -4 & 0 & -4 & 1 \end{array}\right)$$

$$\xrightarrow[R_3 + 3R_2 \to R_3]{} \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 2 & 3 & -4 & 1 \end{array}\right) \xrightarrow[\frac{1}{2}R_3 \to R_3]{} \left(\begin{array}{ccc|ccc} 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{array}\right)$$

$$\xrightarrow[R_1 - 3R_3 \to R_1]{} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{9}{2} & 7 & -\frac{3}{2} \\ 0 & 1 & 2 & 1 & 0 & 0 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{array}\right) \xrightarrow[R_2 - 2R_3 \to R_2]{} \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{9}{2} & 7 & -\frac{3}{2} \\ 0 & 1 & 0 & -2 & -4 & -1 \\ 0 & 0 & 1 & \frac{3}{2} & -2 & \frac{1}{2} \end{array}\right)$$

$$A^{-1} = \begin{pmatrix} -\frac{9}{2} & 7 & -\frac{3}{2} \\ -2 & -4 & -1 \\ \frac{3}{2} & -2 & \frac{1}{2} \end{pmatrix}$$

◊

2.3 Characteristics of Invertible Matrices

Theorem 2.3.1 (Invertible Matrix Theorem)

Let A be an $n \times n$ matrix. The following statements are equivalent:

1) A is invertible.
2) A is row equivalent to I_n.
3) A has n pivot positions.
4) The equation $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
5) The columns of A form a linearly independent set.
6) The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
7) The equation $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b} \in \mathbb{R}^n$.
8) The columns of A span $\mathbb{R}^n$.
9) The linear transformation $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
10) There exists some $n \times n$ matrix C such that $CA = I$.
11) There exists some $n \times n$ matrix D such that $AD = I$.
12) A^{T} is invertible.

Note

In later chapters, we will make multiple amendments to this theorem.

Invertible Linear Transformations

Say the mapping T is defined by $\mathbf{x} \mapsto A\mathbf{x}$. If A is invertible, then we could define an **inverse function** $S = T^{-1}$ as $\mathbf{x} \mapsto A^{-1}\mathbf{x}$. Below, we can show that these mappings are indeed inverses.

$$S(T(\mathbf{x})) = A^{-1}(A\mathbf{x}) = \left(A^{-1}A\right)\mathbf{x} = I\mathbf{x} = \mathbf{x}$$
$$T(S(\mathbf{x})) = A\left(A^{-1}\mathbf{x}\right) = \left(AA^{-1}\right)\mathbf{x} = I\mathbf{x} = \mathbf{x}$$

Theorem 2.3.2

Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear transformation and let A be the standard matrix for T. Then, T is invertible $\Leftrightarrow$ A is invertible. In that case, the linear transformation S defined by $S(\mathbf{x}) = A^{-1}\mathbf{x}$ is a unique mapping that satisfies the following equations:

$$S(T(\mathbf{x})) = \mathbf{x} \quad \forall \mathbf{x} \in \mathbb{R}^n$$
$$T(S(\mathbf{x})) = \mathbf{x} \quad \forall \mathbf{x} \in \mathbb{R}^n$$

2.4 Matrix Factorizations

A factorization of a matrix expresses the matrix as a product of two or more matrices. Often, specific factorizations can make specific matrix computations far less intensive.

LU Factorization

- A **lower triangular matrix** is a square matrix with solely zero entries above the main diagonal.
  - A **unit lower triangular matrix** has only $1s$ along the main diagonal.

Proposition 2.4.1

Products and inverses of unit lower triangular matrices are also unit lower triangular matrices.

- An **upper triangular matrix** is a square matrix with solely zero entries below the main diagonal. The echelon form of an $n \times n$ matrix is an example of this.

If A can be reduced to an echelon form U by a sequence of row replacements corresponding to unit lower triangular matrices $E_1, E_2, \ldots, E_p$, then $A = \left(E_p \cdots E_2 E_1\right)$. Thus, we can factorize A as LU if $L = \left(E_p \cdots E_2 E_1\right)^{-1}$.

Using this factorization to solve $A\mathbf{x} = \mathbf{b}$, we get $(LU)\mathbf{x} = L(U\mathbf{x}) = \mathbf{b}$.

- If we let $U\mathbf{x} = \mathbf{y}$, then we can solve for $\mathbf{x}$ by first solving $L\mathbf{y} = \mathbf{b}$, then substituting into that equation.
- For larger matrices, constructing L and U take far less time than reducing the original matrix to row reduced echelon form.

Algorithm 2.4.2 (LU Factorization)

1) Reduce A to an echelon form U by a sequence of elementary row operations, if possible.
2) Place entries in L such that the *same sequence* of row operations reduces L to I.

For each pivot column in U, let the pivot and the entries below the pivot be a column in L, then apply row replacements to U such that the entries below the pivot become zeros.

- Now, by proposition (2.4.1), L is a unit lower triangular matrix.
- Thus, when copying U's pivot columns onto L, let all entries in each column above the pivot be 0.
- Additionally, divide each column by its pivot.

Note

While L will always be a square matrix, A (and by extension, U) do not have to be square matrices.

Example 2.4.3

Solve $A\mathbf{x} = \mathbf{b}$ using an LU factorization given $A = \begin{pmatrix} 3 & -5 & 3 \\ -9 & 12 & -4 \\ 9 & -12 & 5 \end{pmatrix}$ and $b = \begin{pmatrix} -1 \\ 15 \\ -15 \end{pmatrix}$.

$$\begin{pmatrix} 3 & -5 & 3 \\ -9 & 12 & -4 \\ 9 & -12 & 5 \end{pmatrix} \to \begin{pmatrix} 3 & -5 & 3 \\ 0 & -3 & 5 \\ 0 & 3 & -4 \end{pmatrix} \to \begin{pmatrix} 3 & -5 & 3 \\ 0 & -3 & 5 \\ 0 & 0 & 1 \end{pmatrix} = U$$

$$\begin{pmatrix} 3 & 0 & 0 \\ -9 & -3 & 0 \\ 9 & 3 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 3 & -1 & 1 \end{pmatrix} = L$$

$$L\mathbf{y} = \mathbf{b}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ -3 & 1 & 0 & 15 \\ 3 & -1 & 1 & -15 \end{array}\right) \xrightarrow[R_2+3R_1\to R_2]{} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 12 \\ 3 & -1 & 1 & -15 \end{array}\right) \xrightarrow[R_3-3R_1\to R_3]{} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 12 \\ 0 & -1 & 1 & -12 \end{array}\right)$$

$$\xrightarrow[R_3+R_2\to R_3]{} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 12 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$$\therefore \mathbf{y} = \begin{pmatrix} -1 \\ 12 \\ 0 \end{pmatrix}$$

$$U\mathbf{x} = \mathbf{y}$$

$$\left(\begin{array}{ccc|c} 3 & -5 & 3 & -1 \\ 0 & -3 & 5 & 12 \\ 0 & 0 & 1 & 0 \end{array}\right) \xrightarrow[R_1-3R_3\to R_1]{} \left(\begin{array}{ccc|c} 3 & -5 & 0 & -1 \\ 0 & -3 & 5 & 12 \\ 0 & 0 & 1 & 0 \end{array}\right) \xrightarrow[R_2-5R_3\to R_2]{} \left(\begin{array}{ccc|c} 3 & -5 & 0 & -1 \\ 0 & -3 & 0 & 12 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$$\xrightarrow[-\frac{1}{3}R_2\to R_2]{} \left(\begin{array}{ccc|c} 3 & -5 & 0 & -1 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 0 \end{array}\right) \xrightarrow[R_1+5R_2\to R_1]{} \left(\begin{array}{ccc|c} 3 & 0 & 0 & -21 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$$\xrightarrow[R_1+5R_2\to R_1]{} \left(\begin{array}{ccc|c} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & -4 \\ 0 & 0 & 1 & 0 \end{array}\right)$$

$$\therefore \mathbf{x} = \begin{pmatrix} -7 \\ -4 \\ 0 \end{pmatrix}$$

◇

2.5 Homogeneous Coordinates

The **homogeneous coordinates** of a point $(x, y) \in \mathbb{R}^2$ is $(x, y, 1) \in \mathbb{R}^3$.

- Recall how linear transformations must map $\mathbf{0}$ to $\mathbf{0}$, meaning that we were unable to express translations.
- Using homogeneous coordinates, we can translate vectors by $\begin{pmatrix} h \\ k \end{pmatrix}$ on the plane $z = 1$ using the following standard matrix:

$$A = \begin{pmatrix} 1 & 0 & h \\ 0 & 1 & k \\ 0 & 0 & 1 \end{pmatrix}$$

- The other geometric $\mathbb{R}^2$ linear transformations we covered in (1.9) are expressed with standard matrices similar to their previous ones. For example:

Rotation

$$A = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Reflection over vertical axis

$$A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Homogeneous 3D Coordinates

In general, the homogeneous coordinates of a point $(x, y, z) \in \mathbb{R}^3$ is $\left(\frac{X}{H}, \frac{Y}{H}, \frac{Z}{H}, H\right) \in \mathbb{R}^4$ where $H \neq 0$ and $x = \frac{X}{H}$, $y = \frac{Y}{H}$, and $z = \frac{Z}{H}$.

- Thus, nonzero scalar multiples of $(x, y, z, 1)$ would also be homogeneous coordinates for (x, y, z).

Perspective Projections

Imagine the xy-plane as a viewing frame for a scene, and imagine that a viewer observes from some point $(0, 0, d)$ on the positive z-axis. A **perspective projection** maps each point (x, y, z) onto an *image point* $(x^*, y^*, 0)$ such that those points and the viewing position lie on a single line.

- Drawing out these lines, we find the image's coordinates in terms of the original position and the viewing position.

$$\frac{x^*}{d} = \frac{x}{d - z} \Rightarrow x^* = \frac{dx}{d - z} = \frac{x}{1 - z/d}$$
$$\frac{y^*}{d} = \frac{y}{d - z} \Rightarrow y^* = \frac{dy}{d - z} = \frac{y}{1 - z/d}$$

- Applying homogeneous coordinates, we are mapping $(x, y, z, 1)$ to $\left(\frac{x}{1-z/d}, \frac{y}{1-z/d}, 0, 1\right)$, or equivalently, $\left(x, y, 0, 1 - \frac{z}{d}\right)$.

Now, we can create a general 4×4 matrix $\boldsymbol{P}$ for all $\mathbb{R}^3$ perspective projections.

$$\boldsymbol{P} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{d} & 1 \end{pmatrix}$$

$$\boldsymbol{P}\begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \boldsymbol{P} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & -\frac{1}{d} & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ y \\ 0 \\ 1 - \frac{z}{d} \end{pmatrix} \equiv \begin{pmatrix} \frac{dx}{d-z} \\ \frac{dy}{d-z} \\ 0 \\ 1 \end{pmatrix}$$

3 Determinants

In chapter (2.2), we defined determinants for 2×2 matrices. In this chapter, we will cover the general definition for $n \times n$ matrices.

3.1 Introduction to Determinants

Definition 3.1.1

For $n \geq 2$, the **determinant** of an $n \times n$ matrix $A = (a_{ij})$ is the sum of n terms of the form $\pm a_{1j} \det A_{1j}$.

$$\begin{aligned} \det A &= a_{11} \det A_{11} - a_{12} \det A_{12} + \cdots + (-1)^{1+n} a_{1n} \det A_{1n} \\ &= \sum_{j=1}^{n} (-1)^{1+j} a_{1j} \det A_{1j} \end{aligned}$$

A_{ij} denotes the **(i, j)-minor** of A. That is, the matrix obtained from removing the i^{th} row and j^{th} column from A.

- The determinant of some $n \times n$ matrix $A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix}$ can be expressed using vertical lines in lieu of the parentheses (or brackets).

$$\det A = \begin{vmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{vmatrix}$$

The **(i, j)-cofactor** of some matrix A, denoted C_{ij}, is denoted as follows:

$$C_{ij} = (-1)^{i+j} \det A_{ij}$$

A **cofactor expansion** is a method of calculating a determinant using any row or column. For instance, our definition for determinant uses a *cofactor expansion across the first row*.

Theorem 3.1.2

The determinant of an $n \times n$ matrix A can be computed by a cofactor expansion across any row or down any column.

The cofactor expansion across the i^{th} row of A is

$$\det A = a_{i1} C_{i1} + a_{i2} C_{i2} + \cdots + a_{in} C_{in}$$

The cofactor expansion down the j^{th} column of A is

$$\det A = a_{1j} C_{1j} + a_{2j} C_{2j} + \cdots + a_{nj} C_{nj}$$

- Because we can choose any row or column to do a cofactor expansion along, choosing rows or columns with more zeros is preferable.

Note

The sign of $(-1)^{i+j}$ in every cofactor follows the same checkerboard pattern for every matrix.

$$\begin{pmatrix} + & - & + & \cdots \\ - & + & - & \cdots \\ + & - & + & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

Theorem 3.1.3

If A is triangular or diagonal matrix, then $\det A$ is a product of the entries along the main diagonal of A.

$$\det A = \prod_{i=1}^{n} a_{ii}$$

Proof. Suppose U is an upper triangular matrix. If U is a 2×2 matrix, then

$$\det U = u_{11}u_{22} - u_{12} \cdot 0 = u_{11}u_{22}$$

Now, suppose that if U is $n \times n$, $\det U = u_{11}u_{22}\cdots u_{nn}$. Let U' be an $(n+1) \times (n+1)$ upper triangular matrix such that $U'_{11} = U$. If we do a cofactor expansion down the first column of U', we get:

$$\begin{aligned} \det U' &= u'_{11} \det U'_{11} - 0 \cdot \det U'_{21} + \cdots + 0 \cdot (-1)^{n+1} \det U'_{n1} \\ \det U' &= u'_{11} \det U \\ \det U' &= u'_{11} \left(u_{11}u_{22} \ldots u_{nn} \right) \\ \det U' &= u'_{11}u'_{22}u'_{33} \ldots u'_{(n+1),(n+1)} \end{aligned}$$

It follows that we can apply the same process to a diagonal matrix D, and apply a similar process to a lower triangular matrix L, albeit using the first row of each consecutive minor rather than the first column. □

Important

Because the identity matrix is a diagonal matrix consisting solely of $1s$, for any $n \times n$ identity matrix, $\det I_n = 1$.

3.2 Properties of Determinants

Row Operations

Theorem 3.2.1 (Row Operations)

Let A be an $n \times n$ matrix.

a) If a row replacement is applied to A to produce B, then $\det B = \det A$.
b) If two rows of A are swapped to produce B, then $\det B = -\det A$.
c) If one row of A is scaled by some nonzero constant k, then $\det B = k \det A$.

Note

If an $n \times n$ matrix A is multiplied by some scalar k to produce B, then $\det B = k^n \det A$.

Theorem 3.2.2 (Invertible Matrix Theorem *continued*)

Let A be an $n \times n$ matrix. Then, the following statements are equivalent:

12) *The first twelve statements of the theorem in (2.3).*
13) $\det A \neq 0$

Column Operations

Theorem 3.2.3

If A is an $n \times n$ matrix, then $\det A^{\mathsf{T}} = \det A$.

- Thus, column operations have equivalent effects on determinants as corresponding row operations.

Determinants and Matrix Product

Theorem 3.2.4 (Multiplicative Property)

If A and B are $n \times n$ matrices, then $\det AB = (\det A)(\det B)$.

- So, if we do an LU factorization for some $n \times n$ matrix A, we can find its determinant using $\det L \det U$.

Linearity Property of the Determinant Function

For any $n \times n$ matrix A, $\det A$ is a function of the n column vectors of A. Now, suppose that all columns of A aside from the j^{th} column of A are held fixed such that we can define $\det A$ with transformation $T : \mathbb{R}^n \to \mathbb{R}$ as follows:

$$T(\mathbf{x}) = \det\begin{pmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_{j-1} & \mathbf{x} & \mathbf{a}_{j+1} & \cdots & \mathbf{a}_{n-1} & \mathbf{a}_n\end{pmatrix}$$

This is a linear transformation.

- Because all columns (aside from the j^{th} column) are held fixed, doing cofactor expansions along the j^{th} row shows the linearity properties:

$$\forall \mathbf{u}, \mathbf{v} \in \mathbb{R}^n \quad \text{and} \quad \forall k \in \mathbb{R}$$

$$\begin{aligned}
T(\mathbf{u} + \mathbf{v}) &= (u_1 + v_1)C_{1j} + \cdots + (u_n + v_n)C_{nj} \\
T(\mathbf{u} + \mathbf{v}) &= (u_1C_{1j} + \cdots + u_nC_{nj}) + (v_1C_{1j} + \cdots + v_nC_{nj}) \\
T(\mathbf{u} + \mathbf{v}) &= T(\mathbf{u}) + T(\mathbf{v})
\end{aligned}$$

$$\begin{aligned}
T(k\mathbf{u}) &= ku_1C_{1j} + \cdots + ku_nC_{nj} \\
T(k\mathbf{u}) &= k(u_1C_{1j} + \cdots + u_nC_{nj}) \\
T(k\mathbf{u}) &= kT(\mathbf{u})
\end{aligned}$$

3.3 Cramer's Rule

Cramer's rule is used to solve for singular entries of a solution to $A\mathbf{x} = \mathbf{b}$. Because it involves determinants, its very impractical for hand calculations. Rather, it is very useful for theoretical calculations or small matrices.

Theorem 3.3.1 (Cramer's Rule)

Let A be an $n \times n$ invertible matrix. For any $b \in \mathbb{R}^n$, the unique solution $\mathbf{x}$ of $A\mathbf{x} = \mathbf{b}$ has entries given by

$$x_i = \frac{\det A_i(\mathbf{b})}{\det A}$$

where $1 \le i \le n$ and $A_i(\mathbf{b})$ denotes the matrix obtained from replacing the i^{th} column of A with $\mathbf{b}$.

Proof. Let A be an invertible $n \times n$ matrix with columns $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$.

$$\begin{aligned} A\left(I_i(\mathbf{x})\right) &= A\left(\mathbf{e}_1 \ \cdots \ \mathbf{e}_{i-1} \ \mathbf{x} \ \mathbf{e}_{i+1} \ \cdots \ \mathbf{e}_n\right) \\ A\left(I_i(\mathbf{x})\right) &= \left(A\mathbf{e}_1 \ \cdots \ A\mathbf{e}_{i-1} \ A\mathbf{x} \ A\mathbf{e}_{i+1} \ \cdots \ A\mathbf{e}_n\right) \\ A\left(I_i(\mathbf{x})\right) &= \left(\mathbf{a}_1 \ \cdots \ \mathbf{a}_{i-1} \ \mathbf{b} \ \mathbf{a}_{i+1} \ \cdots \ \mathbf{a}_n\right) \\ A\left(I_i(\mathbf{x})\right) &= A_i(\mathbf{b}) \end{aligned}$$

$$\begin{aligned} \therefore \det\left(AI_i(\mathbf{x})\right) &= \det A_i(\mathbf{b}) \\ \det A \det I_i(\mathbf{x}) &= \det A_i(\mathbf{b}) \\ \det I_i(\mathbf{x}) &= \frac{\det A_i(\mathbf{b})}{\det A} \\ x_i &= \frac{\det A_i(\mathbf{b})}{\det A} \end{aligned}$$

□

A Formula for A^{-1}

Because the product of a matrix and its inverse equals I, the product of a matrix and and the j^{th} column vector of its inverse equals the j^{th} unit basis vector. By Cramer's rule,

$$a_{ij}^{-1} = \frac{\det A_i\left(\mathbf{e}_j\right)}{\det A}$$

Now, we can simplify $\det A_i\left(\mathbf{e}_j\right)$ by cofactor expansion down the i^{th} column, because all entries in the column will be 0 aside from the j^{th} entry.

$$a_{ij}^{-1} = \frac{C_{ji}}{\det A}$$

Thus, if we do this for every entry of A^{-1}, we can create A^{-1}. Factoring out $\frac{1}{\det A}$ from each entry, each (i, j) entry corresponds to cofactor C_{ji} of A. This matrix is known as the **adjugate** of A.

$$\operatorname{adj} A = \begin{pmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \dots & C_{nn} \end{pmatrix}$$

Theorem 3.3.2 (An Inverse Formula)

Let A be an invertible $n \times n$ matrix.

$$A^{-1} = \frac{1}{\det A} \operatorname{adj} A$$

Determinants as Area or Volume

Theorem 3.3.3

If A is a 2×2 matrix, then the area of the parallelogram defined by the columns of A equals $|\det A|$.

If A is a 3×3 matrix, then the volume of the parallelepiped defined by the columns of A equals $|\det A|$.

Note

When using determinants to find area or volume, recall that column vectors should be from the origin. For example, a side between points $(1, 1)$ and $(5, 1)$ would correspond to the vector $\binom{4}{0}$.

- We can interpret the determinant of a matrix as how much vectors are scaled under a corresponding transformation.

4 Vector Spaces

4.1 Vector Spaces and Subspaces

We have worked with n-dimensional vector spaces $\mathbb{R}^n$, but the definition of a vector space allows us to classify many different systems as vector spaces as well.

Definition 4.1.1

A **vector space** is a nonempty set V of elements known as *vectors* with two defined operations: *addition* and *scalar multiplication*. These operations are subject to to the following ten axioms:

$$\forall \mathbf{u}, \mathbf{v}, \mathbf{w} \in V \quad \text{and} \quad \forall \text{ scalars } c, d$$

1) $\mathbf{u} + \mathbf{v} \in V$
2) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
3) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
4) There exists a zero vector $\mathbf{0}$ such that $\mathbf{u} + \mathbf{0} = u$
5) For each $\mathbf{u} \in V$, there exists a vector $-\mathbf{u} \in V$ such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$
6) $c\mathbf{u} \in V$
7) $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
8) $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$
9) $c(d\mathbf{u}) = (cd)\mathbf{u}$
10) $1\mathbf{u} = \mathbf{u}$

♣

- You might recognize many of these axioms from subchapter (1.3).
- Plenty of common vector spaces include the set of $n \times n$ matrices, denoted $M_{n\times n}$, and the set of polynomials up to degree n, denoted $\mathbb{P}_n$.

Subspaces

Many times, a vector space will be a subset of some larger vector space. In this case, we only need to check three out of the ten axioms to prove that it is a vector space because the other axioms will be *satisfied by default.*

Definition 4.1.2

A **subspace** of a vector space V is a subset H of V with the following three properties:
1) The zero vector of V is in H.
2) H is closed under addition.
3) H is closed under scalar multiplication.

♣

- All subspaces are vector spaces, and like subsets, all vector spaces are subspaces of themselves.
- The smallest subspace of any vector space V is the **zero subspace**, $\{\mathbf{0}\}$.

Spanning Sets and Subspaces

Often, subspaces will be expressed using a **spanning set**. In this form, we can use previously defined axioms for vector spaces and linear combinations to prove that they are indeed subspaces of a particular vector space.

Theorem 4.1.3

If $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p$ are in a vector space V, then $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\}$ is a subspace of V.

4.2 Special Subspaces

Null Space

Definition 4.2.1

The **null space** of an $m \times n$ matrix, denoted Nul A, is the set of all solutions of the homogeneous equation $A\mathbf{x} = \mathbf{0}$.

$$\text{Nul } A = \{\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n, A\mathbf{x} = \mathbf{0}\}$$

- Essentially, given some $m \times n$ matrix A, Nul A is the set of all vectors in $\mathbb{R}^n$ that are mapped to the zero vector in $\mathbb{R}^m$.
- Additionally, the vectors from the parametric vector form of the solution set to $A\mathbf{x} = \mathbf{0}$ are the vectors that make up the spanning set for Nul A.

Theorem 4.2.2

The null space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$.

Column Space

Definition 4.2.3

The **column space** of an $m \times n$ matrix A, denoted Col A, is the set of all linear combinations of the columns of A. If the columns of A are $\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n$, then

$$\text{Col } A = \text{Span}\{\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n\}$$

Theorem 4.2.4

The column space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^m$.

Row Space

Definition 4.2.5

The **row space** of an $m \times n$ matrix A, denoted Row A, or equivalently, Col A^{T}, is the set of all linear combinations of the rows of A.

Theorem 4.2.6

The row space of an $m \times n$ matrix A is a subspace of $\mathbb{R}^n$.

Kernel and Range of a Linear Transformation

Subspaces of vectors spaces other than $\mathbb{R}^n$ are typically defined using a linear transformation rather than a matrix. Thus, we will generalize our linear transformation definition from subchapter (1.8).

Definition 4.2.7

A **linear transformation** T from a vector space V into a vector space W is a mapping that assigns each vector $\mathbf{x} \in V$ to a unique vector $T(\mathbf{x}) \in W$ such that

i) $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})\ \forall \mathbf{u}, \mathbf{v} \in V$.

ii) $T(c\mathbf{u}) = cT(\mathbf{u})\ \forall$ scalars c and $\forall \mathbf{u} \in V$.

♣

The **kernel** of a linear transformation T from a vector space V to a vector space W, denoted $\ker T$, is the set of all vectors in V that map to the zero vector of U.

$$\ker T = \{\mathbf{u} \mid \mathbf{u} \in V, T(\mathbf{u}) = \mathbf{0}\}$$

- If T is associated with a standard matrix A, then $\ker T = \operatorname{Nul} A$.
- Similarly, $\ker T$ is a subspace of V.

Meanwhile, the **range** of T is the set of all vectors in W of the form $T(\mathbf{x})$.

- If T is associated with a standard matrix A, then the range of T equals $\operatorname{Col} A$.
- Similarly, the range of T is a subspace of W.

4.3 Linearly Independent Sets and Bases

Definition 4.3.1

Let V be a vector space and let $\mathcal{B}$ be a subset of V. $\mathcal{B}$ is a **basis** for V if the following is true:

1) $\mathcal{B}$ is a linearly independent set.
2) The subspace spanned by $\mathcal{B}$ coincides with V, that is,

$$V = \text{Span } \mathcal{B}$$

The **standard basis** for $\mathbb{R}^n$ is defined as follows:

$$\{\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n\}$$

The Spanning Set Theorem

Theorem 4.3.2 (The Spanning Set Theorem)

Let $S = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ be subset of vector space V, and let $H = \text{Span } S$.

a) If a vector $\mathbf{v}_k \in S$ is a linear combination of the other vectors in S, then the set formed by removing $\mathbf{v}_k$ still spans H.
b) If $H \neq \{\mathbf{0}\}$, some subset of S is a basis for H.

- When solving a homogeneous system $A\mathbf{x} = \mathbf{0}$ for Nul A, the spanning set found from the solution set is usually a basis for Nul A because the vectors will always be linearly independent unless $\mathbf{0}$ is in the spanning set.
- Meanwhile, the columns of some matrix A are not guaranteed to be linearly independent. A good way to form a basis for Col A is to only use the pivot columns of A. This works because the other columns of A are some linear combination of the pivot columns.

Caution

When finding the pivot columns of A using row reduction, do not use the columns of the reduced matrix.

- Similarly, the pivot rows of A, or equivalently, the pivot columns of A^T also form a basis for Row A. Unlike with the column space, we can use the pivot rows of the reduced A or the pivot columns of the reduced Col A^T in our basis.

Note

A basis is essentially the *smallest possible spanning set* for a vector space.

4.4 Coordinate Systems

Theorem 4.4.1 (The Unique Representation Theorem)

Let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_p\}$ be a basis for a vector space V. For each vector $\mathbf{x} \in V$, there exists a unique set of scalars $c_1, c_2, \ldots, c_p$ such that

$$\mathbf{x} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \cdots + c_p\mathbf{b}_p$$

Proof. Suppose $\mathbf{x}$ is a vector in V and $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_p\}$ is a basis for V such that

$$\mathbf{x} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \cdots + c_p\mathbf{b}_p$$

Now, suppose that $\mathbf{x}$ has another representation using the vectors of $\mathcal{B}$:

$$\mathbf{x} = d_1\mathbf{b}_1 + d_2\mathbf{b}_2 + \cdots + d_p\mathbf{b}_p$$

where $d_1, d_2, \ldots, d_p$ is another set of scalars.

$$\begin{aligned} \mathbf{x} - \mathbf{x} &= (c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \cdots + c_p\mathbf{b}_p) - (d_1\mathbf{b}_1 + d_2\mathbf{b}_2 + \cdots + d_p\mathbf{b}_p) \\ \mathbf{0} &= (c_1 - d_1)\mathbf{b}_1 + (c_2 - d_2)\mathbf{b}_2 + \cdots + (c_p - d_p)\mathbf{b}_p \end{aligned}$$

By definition, $\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_p$ are linearly independent. Thus, for each j such that $1 \le j \le p$:

$$\begin{aligned} c_j - d_j &= 0 \\ c_j &= d_j \end{aligned}$$

□

Definition 4.4.2

Suppose $\mathcal{B} = \{\mathbf{b}_1,, \mathbf{b}_2, \ldots, \mathbf{b}_p\}$ is a basis for a vector space V and $\mathbf{x} \in V$. The **coordinates of x relative to $\mathcal{B}$**, also known as the **$\mathcal{B}$-coordinates of x**, are the scalars $c_1, c_2, \ldots, c_p$ such that $\mathbf{x} = c_1\mathbf{b}_1 + c_2\mathbf{b}_2 + \cdots + c_p\mathbf{b}_p$.

The following vector is the **coordinate vector of x relative to $\mathcal{B}$** or the **$\mathcal{B}$-coordinate vector of x**.

$$(\mathbf{x})_{\mathcal{B}} = \begin{pmatrix} c_1 \\ \vdots \\ c_p \end{pmatrix}$$

Graphical Interpretation

We can interpret plotting points onto *normal* graphing paper as a coordinate mapping from a set of points to $\mathbb{R}^2$ *relative to the standard basis* $\{\mathbf{e}_1, \mathbf{e}_2\} = \left\{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}\right\}$ because the standard axes are perpendicular to each other.

- Thus, plotting points relative to another basis would *warp the axes.*

The Coordinate Mapping

Often, finding bases for vector spaces different from $\mathbb{R}^n$ can be difficult due to their form. We can use a coordinate mapping to $\mathbb{R}^n$ to covert them to a more familiar form.

Theorem 4.4.3

Let $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_p\}$ be a basis for a vector space V. Then, the **coordinate mapping** $\mathbf{x} \mapsto (\mathbf{x})_{\mathcal{B}}$ is a bijective linear transformation from V to $\mathbb{R}^n$.

- This mapping is an **isomorphism** from V onto $\mathbb{R}^n$, as even with potential differences in notation, they are generally equivalent in terms of vector spaces.

Example 4.4.4

Is the set of vectors $\{t^2+t+1, t^2+2t+3, t^2+3t+1\} \in \mathbb{P}_2$ linearly independent or dependent?

Let $T : \mathbb{P}_2 \to \mathbb{R}^3$ be a coordinate mapping defined as follows:

$$at^2 + bt + c \mapsto \begin{pmatrix} a \\ b \\ c \end{pmatrix} \quad \text{for all scalars } a, b, c$$

Now, let $T(t^2+t+1)$, $T(t^2+2t+3)$, and $T(t^2+3t+1)$ be the columns of A.

$$A = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 1 \end{pmatrix} \xrightarrow[R_2 - R_1 \to R_2]{} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & 3 & 1 \end{pmatrix} \xrightarrow[R_3 - R_1 \to R_3]{} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 2 & 0 \end{pmatrix} \xrightarrow[R_2 \leftrightarrow R_3]{} \begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix} \xrightarrow[R_1 - \frac{1}{2}R_2 \to R_1]{} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 1 & 2 \end{pmatrix}$$

$$\xrightarrow[R_3 - \frac{1}{2}R_2 \to R_3]{} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \xrightarrow[R_1 - \frac{1}{2}R_3 \to R_1]{} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix} \xrightarrow[\frac{1}{2}R_2 \to R_2]{} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix} \xrightarrow[\frac{1}{2}R_3 \to R_3]{} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

By the Invertible Matrix Theorem, the columns of A, $T(t^2+t+1)$, $T(t^2+2t+3)$, and $T(t^2+3t+1)$, are linearly independent. Because T is a bijective coordinate mapping, $\{t^2+t+1, t^2+2t+3, t^2+3t+1\}$ is a linearly independent set.

4.5 Dimension of a Vector Space

Theorem 4.5.1

If a vector space V has a basis $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n\}$, then any subset of V with more than n vectors is linearly dependent.

Theorem 4.5.2

If a vector space V has a basis of n vectors, then every basis of V must contain n vectors.

Definition 4.5.3

The **dimension** of a vector space V, denoted $\dim V$, is the number of vectors in a basis for V.

Note

$\dim\{\mathbf{0}\} = 0$ because the only basis for the zero subspace is the empty set, $\varnothing$.

- Now, it follows that the dimension of $\mathbb{R}^n$ is n, as the standard basis has n basis vectors.

Rank and Nullity

Definition 4.5.4

The **rank** of an $m \times n$ matrix A is equal to $\dim(\operatorname{Col} A)$.

- In other words, the rank of an $m \times n$ matrix A is the *number of pivot columns*.
- By extension, the dimension of Row A is equal to the rank of A.

Definition 4.5.5

The **nullity** of an $m \times n$ matrix A is equal to $\dim(\operatorname{Nul} A)$.

- In other words, the nullity of an $m \times n$ matrix A is the *number of free variables*.

Theorem 4.5.6 (The Rank Theorem)

Given an $m \times n$ matrix A

$$\operatorname{rank} A + \operatorname{nullity} A = \text{number of columns in } A$$

Invertible Matrix Theorem Amendments

Theorem 4.5.7 (Invertible Matrix Theorem *continued*)

Let A be an $n \times n$ matrix. Then, the following statements are equivalent:

13) *The first thirteen statements of the theorem in (2.3) and (3.2).*
14) The columns of A form a basis for $\mathbb{R}^n$.
15) $\text{Col}\, A = \mathbb{R}^n$
16) $\text{rank}\, A = n$
17) $\text{nullity}\, A = 0$
18) $\text{Nul}\, A = \{\mathbf{0}\}$

4.6 Change of Basis

son

4.7 Digital Signal Processing

A **signal** is an infinite sequence of numbers. The set of all signals, denoted $\mathbb{S}$, is a vector space. While infinite, it is possible to test their linear independence using just a few terms. This will be covered in the next chapter, as signals will be the general solution to difference equations.

4.8 Applications to Difference Equations

Linear Independence of Signals

Say we want to test the linear independence of three signals: $\{u_k\}$, $\{v_k\}$, and $\{w_k\}$. Then, the following equation should imply $c_1 = c_2 = c_3 = 0$.

$$c_1 u_k + c_2 v_k + c_3 w_k = 0 \quad \forall k \geq 0$$

Now, lets say they do. Then, the equation implies that they are satisfied for k, $k+1$, and $k+2$ for any $k \in \mathbb{N}$. We can test this using a **Casorati matrix**, a coefficient matrix containing consecutive terms of multiple signals:

$$\begin{pmatrix} u_k & v_k & w_k \\ u_{k+1} & v_{k+1} & w_{k+1} \\ u_{k+2} & v_{k+2} & w_{k+2} \end{pmatrix}$$

If its columns are linearly independent, then the signals are linearly independent.

Linear Difference Equations

A **linear difference equation** or **linear recurrence relation** of order n is defined as follows:

$$a_0 y_{k+n} + a_1 y_{k+n-1} + \cdots + a_{n-1} y_{k+1} + a_n y_k = z_k \quad \forall k \geq 0$$

where $a_0, a_1, \ldots, a_{n-1}, a_n$ are scalars (with a_0 and a_n nonzero) and z_k is some signal. Generally, $a_0 = 0$.

- If $\{z_k\}$ is the *zero sequence*, then the difference equation is **homogeneous**.
- Otherwise, the difference equation is **nonhomogeneous**.

The general solution of a homogeneous difference equation is generally $\{y_k\} = \{r^k\}$. We can solve for r using the **auxiliary equation**, which can be further simplified into the **auxiliary polynomial**:

$$a_0 r^{k+n} + a_1 r^{k+n-1} + \cdots + a_{n-1} r^{k+1} + a_n r^k = 0$$
$$a_0 r^n + a_1 r^{n-1} + \cdots + a_{n-1} r^1 + a_n = 0$$

The general solution of a nonhomogeneous difference equation requires us to solve for some constant solution (let $y_k, y_{k+1}, \ldots, y_{k+n}$ equal some variable c), then add it to the corresponding general homogeneous solution.

Example 4.8.1

What is the general solution to the following recurrence relation?

$$y_{k+1} - 2y_k = 1$$

Constant Solution

$$\begin{aligned} c - 2c &= 1 \\ -c &= 1 \\ c &= -1 \end{aligned}$$

Corresponding Homogeneous Solution

$$\begin{aligned} r^{k+1} - 2r^k &= 0 \\ r - 2 &= 0 \\ r &= 2 \end{aligned}$$

$$\therefore \left\{c_1 2^k - 1\right\}$$

Example 4.8.2

\$150,000 is borrowed at 6% interest compounded monthly. Monthly payments are \$1000 dollars. Solve the difference equation for an explicit formula for the balance due after k months.

$$\begin{aligned} y_k &= \left(1 + \frac{0.06}{12}\right) y_{k-1} - 1000 \\ y_k &= (1.005) y_{k-1} - 1000 \end{aligned}$$

Constant Solution

$$\begin{aligned} c &= 1.005c - 1000 \\ -0.005c &= -1000 \\ c &= 200000 \end{aligned}$$

Initial Value

$$\begin{aligned} y_k &= a_0(1.005)^k + 200000 \\ \Rightarrow 150000 &= a_0(1.005)^0 + 200000 \\ a_0 &= -50000 \end{aligned}$$

Explicit Formula

$$\therefore y_k = 200000 - 50000(1.005)^k$$

Theorem 4.8.3

The set H of all solutions to the n^{th} order homogeneous linear difference equation

$$y_{k+n} + a_1 y_{k+n-1} + \cdots + a_{n-1} y_{k+1} + a_n y_k = 0 \quad \forall k \geq 0$$

is an n-dimensional vector space.

5 Eigen-everything

5.1 Eigenvectors and Eigenvalues

Definition 5.1.1 (Eigenvector and Eigenvalue)

An **eigenvector** of an $n \times n$ matrix A is any nonzero vector $\mathbf{x}$ such that $A\mathbf{x} = \lambda\mathbf{x}$ for some scalar λ. A scalar λ is an **eigenvalue** of A if there exists a nontrivial solution $\mathbf{x}$ to $A\mathbf{x} = \lambda\mathbf{x}$ such that $\mathbf{x}$ is the eigenvector corresponding to λ.

♣

- An **eigenspace** consists of all eigenvectors corresponding to some eigenvalue λ, and forms a subspace of $\mathbb{R}^n$.
- Geometrically, an eigenvalue is a scaling factor in a specific direction during a linear transformation.

ⓘ Note

While eigenvectors cannot be zero, *eigenvalues can be zero.* However, if that is the case, then $A\mathbf{x} = 0\mathbf{x}$, or equivalently, $A\mathbf{x} = \mathbf{0}$, has a nontrivial solution, meaning A *is not invertible.*

5.2 The Characteristic Equation

To solve for the eigenvectors, and by extension, the eigenvalues of a matrix, we can take advantage of the general matrix equation for eigenvectors.

$$\begin{aligned} A\mathbf{x} &= \lambda\mathbf{x} \\ A\mathbf{x} - \lambda\mathbf{x} &= 0 \\ (A - \lambda I_n)\mathbf{x} &= 0 \end{aligned}$$

- Now, by definition, the homogeneous equation above must have a nontrivial solution.
- Then, by the Invertible Matrix Theorem, $(A - \lambda I_n)$ is not invertible. Equivalently, its determinant is zero. This gives us the **characteristic equation**, a n^{th} degree polynomial whose roots are A's eigenvalues.

Definition 5.2.1 (Characteristic Equation)

Let A be some $n \times n$ matrix. The characteristic equation is defined as follows:

$$\det(A - \lambda I_n) = 0$$

A scalar λ may only be an eigenvalue for A if it satisfies the equation above. ♣

Proposition 5.2.2

If A is an $n \times n$ matrix, then A^{T} has the same eigenvalues. ♠

Proof. Suppose A and A^{T} are some $n \times n$ matrices whose eigenvalues are given by $\det(A - \lambda I)$ and $\det(A^{\mathsf{T}} - \lambda I)$, respectively.

$$\begin{aligned} \det(A^{\mathsf{T}} - \lambda I) &= \det(A^{\mathsf{T}} - \lambda I^{\mathsf{T}}) \\ \det(A^{\mathsf{T}} - \lambda I) &= \det(A - \lambda I) \end{aligned}$$ □

Example 5.2.3

Solve for the eigenspace(s) of $A = \begin{pmatrix} -1 & 1 & 2 \\ -2 & 2 & 2 \\ -2 & 1 & 3 \end{pmatrix}$.

$$\det(A - \lambda I) = 0 \Rightarrow \lambda = 1, 1, 2$$

$$\lambda_1 = 1$$

$$A - \lambda_1 I = A - I$$

$$\begin{pmatrix} -2 & 1 & 2 \\ -2 & 1 & 2 \\ -2 & 1 & 2 \end{pmatrix} \xrightarrow[\text{rref}]{} \begin{pmatrix} 1 & -\frac{1}{2} & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\therefore \Lambda_1 = \left\{ \begin{pmatrix} \frac{1}{2} \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \right\}$$

$$\lambda_2 = 2$$

$$A - \lambda_2 I = A - 2I$$

$$\begin{pmatrix} -3 & 1 & 2 \\ -2 & -1 & 2 \\ -2 & 1 & 0 \end{pmatrix} \xrightarrow[\text{rref}]{} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\therefore \Lambda_2 = \left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}$$

◊

Similarity

If A and B are $n \times n$ matrices, then A is **similar to** B if there exists an invertible matrix P such that $A = PBP^{-1}$, or equivalently, $B = P^{-1}AP$.

- This relationship is *symmetric*.
- The mapping $A \mapsto P^{-1}AP$ is known as a **similarity transformation**.

Theorem 5.2.4

If two $n \times n$ matrices A and B are similar, then their characteristic polynomials, and by extension, their eigenvalues, are the same.

Proof. Let P be some $n \times n$ invertible matrix.

$$\begin{aligned} A - \lambda I &= PBP^{-1} - \lambda PP^{-1} \\ A - \lambda I &= P(B - \lambda)P^{-1} \\ \det(A - \lambda I) &= \det\left(P(B - \lambda)P^{-1}\right) \\ \det(A - \lambda I) &= \det\left(P(B - \lambda)P^{-1}\right) \\ \det(A - \lambda I) &= \det(P)\det(B - \lambda)\det\left(P^{-1}\right) \\ \det(A - \lambda I) &= \det(P)\det(B - \lambda)\left(\frac{1}{\det(P)}\right) \\ \det(A - \lambda I) &= \det(B - \lambda) \end{aligned}$$

□

5.3 Diagonalization

A matrix is **diagonalizable** if it is similar to a diagonal matrix.

Theorem 5.3.1 (The Diagonalization Theorem)

An $n \times n$ matrix A is diagonalizable $\Leftrightarrow$ A has n linearly independent eigenvectors.

Specifically, $A = PDP^{-1}$ given diagonal matrix $D \Leftrightarrow$ the columns of P are n linearly independent eigenvectors of A such that each i^{th} column of P corresponds to each i^{th} eigenvalue along D's main diagonal.

- In other words, A is diagonalizable $\Leftrightarrow$ there exists enough eigenvectors to form an **eigenvector basis** of $\mathbb{R}^n$.

Proof. Let P be the a matrix whose columns are A's eigenvectors, $\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n$ and let D be a diagonal matrix whose diagonal entries are A's eigenvalues $\lambda_1, \lambda_2, \ldots, \lambda_n$. Since the eigenvectors of A are linearly independent, the columns of P are linearly independent. By the Invertible Matrix Theorem, P is invertible. Since P, D, and P^{-1} are all well-defined, the diagonalization PDP^{-1} exists as follows:

$$\begin{aligned}
AP &= A\begin{pmatrix}\mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n\end{pmatrix} \\
AP &= \begin{pmatrix}A\mathbf{v}_1 & A\mathbf{v}_2 & \cdots & A\mathbf{v}_n\end{pmatrix} \\
AP &= \begin{pmatrix}\lambda_1\mathbf{v}_1 & \lambda_2\mathbf{v}_2 & \cdots & \lambda_n\mathbf{v}_n\end{pmatrix} \\
APP^{-1} &= \begin{pmatrix}\lambda_1\mathbf{v}_1 & \lambda_2\mathbf{v}_2 & \cdots & \lambda_n\mathbf{v}_n\end{pmatrix}P^{-1} \\
A &= \begin{pmatrix}\lambda_1\mathbf{v}_1 & \lambda_2\mathbf{v}_2 & \cdots & \lambda_n\mathbf{v}_n\end{pmatrix}P^{-1} \\
A &= PDP^{-1}
\end{aligned}$$

Now, suppose A is diagonalizable such that $A = PDP^{-1}$. This is equivalent to $AP = PD$. Now, let P's columns be vectors $\mathbf{v}_1, \mathbf{v}_2, \cdots, \mathbf{v}_n$ and let D's diagonal entries be $\lambda_1, \lambda_2, \ldots, \lambda_n$.

$$\begin{aligned}
AP &= PD \\
A\begin{pmatrix}\mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n\end{pmatrix} &= \begin{pmatrix}\mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n\end{pmatrix}\begin{pmatrix}\lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \ldots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n\end{pmatrix} \\
\begin{pmatrix}A\mathbf{v}_1 & A\mathbf{v}_2 & \cdots & A\mathbf{v}_n\end{pmatrix} &= \begin{pmatrix}\lambda_1\mathbf{v}_1 & \lambda_2\mathbf{v}_2 & \cdots & \lambda_n\mathbf{v}_n\end{pmatrix}
\end{aligned}$$

□

- This factorization is particularly helpful for calculating large k^{th} powers of matrices because raising a diagonal matrix to the k^{th} power is the same as raising their diagonal entries to the k^{th} power. In fact:

$$A^k = \left(PDP^{-1}\right)^k = \left(PDP^{-1}\right)\left(PDP^{-1}\right)\cdots = PD^kP^{-1}$$

Multiplicity

As shown prior, its possible to get a characteristic equation with **multiplicity**. That is, more than one root with the same value. But, diagonalization does not require *n distinct* eigenvalues.

Theorem 5.3.2

Let A be an $n \times n$ matrix with distinct eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_p$ and corresponding eigenspaces $\Lambda_1, \Lambda_2, \dots, \Lambda_p$.

1) $\dim \Lambda_k \leq$ multiplicity of λ_k for each k^{th} eigenspace and eigenvalue.
2) A is diagonalizable $\Leftrightarrow \dim \Lambda_1 + \dim \Lambda_2 + \dots + \dim \Lambda_p = n$.
   a) Essentially, $\dim \Lambda_k =$ multiplicity of λ_k.
3) If A is diagonalizable and $\mathcal{B}_k$ is the basis associated with each k^{th} eigenspace Λ_k, then $\Lambda_1 \cup \Lambda_2 \cup \dots \cup \Lambda_k$ forms an **eigenvector basis** for $\mathbb{R}^n$.

5.4 Complex Eigenvalues

Complex roots of the characteristic equation have real-life applications, generally representing rotations or periodic motion. In fact, the rotational matrices we covered in (1.9) have complex eigenvalues for almost all values of θ.

A **complex eigenvalue** λ corresponding to a **complex eigenvector** $\mathbf{x}$ still must satisfy both $\det(A - \lambda I) = 0$ and $A\mathbf{x} = \lambda\mathbf{x}$.

- Complex eigenvalues always come in conjugate pairs: $\alpha \pm \beta i$.
- Often, we split complex eigenvectors into their *real* and *imaginary* parts. This will be important when studying their relation to rotations.

$$\mathbf{x} = \operatorname{Re}\mathbf{x} + \operatorname{Im}\mathbf{x}$$

Example 5.4.1

Find the eigenvalues and eigenvectors of $A = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$.

$$\begin{aligned} \det(A - \lambda I) = (1 - \lambda)(3 - \lambda) - (2)(-1) &= 0 \\ \lambda^2 - 4\lambda + 3 + 2 &= 0 \\ \lambda^2 - 4\lambda + 5 &= 0 \end{aligned}$$

$$\begin{aligned} \lambda &= \frac{4 \pm \sqrt{(-4)^2 - 4(5)}}{2} \\ \lambda &= \frac{4 \pm 2i}{2} \\ \lambda &= 2 \pm i \end{aligned}$$

$$\lambda_1 = 2 + i \qquad\qquad \lambda_2 = 2 - i$$

$$\begin{aligned} \big(1 - (2 + i)\big)x_1 + 2x_2 &= 0 \\ -x_1 + \big(3 - (2 + i)\big)x_1 &= 0 \end{aligned} \qquad\qquad \begin{aligned} \big(1 - (2 - i)\big)x_1 + 2x_2 &= 0 \\ -x_1 + \big(3 - (2 - i)\big)x_1 &= 0 \end{aligned}$$

$$\begin{aligned} (-1 - i)x_1 + 2x_2 &= 0 \\ -x_1 + (1 - i)x_1 &= 0 \end{aligned} \qquad\qquad \begin{aligned} (-1 + i)x_1 + 2x_2 &= 0 \\ -x_1 + (1 + i)x_1 &= 0 \end{aligned}$$

$$\therefore \mathbf{v}_1 = \begin{pmatrix} 1 - i \\ 1 \end{pmatrix} \qquad\qquad \therefore \mathbf{v}_1 = \begin{pmatrix} 1 + i \\ 1 \end{pmatrix}$$

◊

Rotational Factorization

While we cannot diagonalize matrices with complex eigenvalues, we can still factorize them. In fact, they are similar to a rotational matrix, as covered in the following theorem.

Theorem 5.4.2

Let A be a real 2×2 matrix with complex eigenvalue $\lambda = \alpha - \beta i$ for $\beta \neq 0$ and an associated eigenvector $\mathbf{v} \in \mathbb{C}^2$.

$$A = PCP^{-1}$$

where

$$P = (\operatorname{Re}\mathbf{v} \;\; \operatorname{Im}\mathbf{v}) \quad \text{and} \quad C = \begin{pmatrix} \alpha & -\beta \\ \beta & \alpha \end{pmatrix}$$

5.5 Discrete Dynamical Systems

An application of eigenvalues and eigenvectors is using them to understand dynamical systems given by the equation $\mathbf{x}_{k+1} = A\mathbf{x}_k$. We used this model in (1.10) and will apply it further in (5.9).

- Now, we can generalize this equation for some k^{th} vector as follows:

$$\begin{aligned}
\mathbf{x}_k &= A\mathbf{x}_{k-1} \\
\mathbf{x}_k &= A\Big(\cdots\big(A\mathbf{x}_0\big)\Big) \\
\mathbf{x}_k &= A^k\mathbf{x}_0 \\
\mathbf{x}_k &= c_1A^k\mathbf{v}_1 + \cdots + c_nA^k\mathbf{v}_n \\
\mathbf{x}_k &= c_1\lambda_1^k\mathbf{v}_1 + \cdots + c_n\lambda_n^k\mathbf{v}_n
\end{aligned}$$

The equation above is the **eigenvector decomposition** for $\mathbf{x}_k$.

- This decomposition can tell us a lot about the system in the long run, as for any i^{th} eigenvalue, if $|\lambda_i| < 1$, then $c_i\lambda_i^k\mathbf{v}_i = 0$ as $k \to \infty$.
- This works even for complex eigenvalues. For any complex eigenvalue $\lambda = \alpha \pm \beta$, we simply check if its **modulus** is less than 1. That is, $\|\lambda\| = \sqrt{\alpha^2 + \beta^2} < 1$

Predator-Prey Systems

A common discrete system involves a vector of two populations. Both populations grow based on their current population. However, the **prey** population grows slower or even decays the greater the other population is. Meanwhile, the **predator** population grows faster (or decays slower) the greater the other population is.

In general:

$$\begin{aligned}
x_{k+1} &= c_1x_k - c_2y_k \\
y_{k+1} &= d_1x_k + d_2y_k
\end{aligned} \qquad \begin{pmatrix} x_{k+1} \\ y_{k+1} \end{pmatrix} = A^k \begin{pmatrix} x_0 \\ y_0 \end{pmatrix}$$

where x represents the prey population, y represents the predator population, and c_1, c_2, d_1, d_2 are some positive real numbers.

Example 5.5.1

In the following predator-prey system, the prey population x and predator population y are defined as follows:

$$\begin{aligned} x_{k+1} &= 1.175x_k - 0.375y_k \\ y_{k+1} &= 0.125x_k + 0.675y_k \end{aligned}$$

What can be said about the populations in the long run? What would be an explicit equation for $\begin{pmatrix} x_k \\ y_k \end{pmatrix}$?

$$A = \begin{pmatrix} 1.175 & -0.375 \\ 0.125 & 0.675 \end{pmatrix}$$

$$\det(A - \lambda I) = 0 \Rightarrow \lambda = 0.8, 1.05$$

$$\lambda_1 = 0.8$$

$$\begin{aligned} (1.175 - 0.8)x_0 - 0.375y_0 &= 0 \\ 0.125x_0 + (0.675 - 0.8)y_0 &= 0 \end{aligned}$$

$$\begin{aligned} (0.375)x_0 - 0.375y_0 &= 0 \\ 0.125x_0 + (-0.125)y_0 &= 0 \end{aligned}$$

$$\therefore \mathbf{v}_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\lambda_2 = 1.05$$

$$\begin{aligned} (1.175 - 1.05)x_0 - 0.375y_0 &= 0 \\ 0.125x_0 + (0.675 - 1.05)y_0 &= 0 \end{aligned}$$

$$\begin{aligned} (0.125)x_0 - 0.375y_0 &= 0 \\ 0.125x_0 + (-0.375)y_0 &= 0 \end{aligned}$$

$$\therefore \mathbf{v}_1 = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

In almost all scenarios, the prey population will be roughly three times the predator population and the total population will grow by approximately 5% per year. However, any initial states with equal prey and predator populations will decay by 20% per year.

$$P = \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix} \qquad D = \begin{pmatrix} 1.05 & 0 \\ 0 & 0.8 \end{pmatrix}$$

$$\begin{aligned} A^k &= PD^kP^{-1} \\ A^k &= \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1.05 & 0 \\ 0 & 0.8 \end{pmatrix}^k \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix}^{-1} \\ A^k &= \begin{pmatrix} 3 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1.05^k & 0 \\ 0 & 0.8^k \end{pmatrix} \begin{pmatrix} \frac{1}{2} & -\frac{1}{2} \\ -\frac{1}{2} & \frac{3}{2} \end{pmatrix} \\ A^k &= \begin{pmatrix} 1.5(1.05)^k - 0.5(0.8)^k & 1.5(0.8)^k - 1.5(1.05)^k \\ 0.5(1.05)^k - 0.5(0.8)^k & 1.5(0.8)^k - 0.5(1.05)^k \end{pmatrix} \end{aligned}$$

$$\begin{pmatrix} x_k \\ y_k \end{pmatrix} = (1.05)^k \begin{pmatrix} 1.5x_0 - 1.5y_0 \\ 0.5x_0 - 0.5y_0 \end{pmatrix} + (0.8)^k \begin{pmatrix} -0.5x_0 + 1.5y_0 \\ -0.5x_0 + 1.5y_0 \end{pmatrix}$$

5.6 Applications to Differential Equations

Differential equations relate functions to their derivatives. Because differentiation of functions is linear, systems of differential equations *are linear systems*. Unlike difference equations which model sequential steps, differential equations model *continuous change*.

- Below, $x_1, x_2, \ldots, x_n$ are functions of time t.

$$\begin{aligned} x_1' &= a_{11}x_1 + \cdots + a_{1n}x_n \\ x_2' &= a_{21}x_1 + \cdots + a_{2n}x_n \\ &\vdots \\ x_n' &= a_{n1}x_1 + \cdots + a_{nn}x_n \end{aligned} \qquad \begin{gathered} \mathbf{x}'(t) = A\mathbf{x}(t) \\ \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix}' = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} x_1(t) \\ \vdots \\ x_n(t) \end{pmatrix} \end{gathered}$$

A system is **decoupled** when the derivative of each function only depends on itself. If A is a diagonal matrix, then the system is decoupled.

$$\begin{pmatrix} x_1'(t) \\ x_2'(t) \end{pmatrix} = \begin{pmatrix} c & 0 \\ 0 & d \end{pmatrix} \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} \Rightarrow \begin{aligned} x_1'(t) = cx_1(t) \\ x_2'(t) = cx_2(t) \end{aligned}$$

- Now, by calculus, the solutions, called **eigenfunctions** in linear algebra, are k_1e^{ct} and k_2e^{dt} for any constant numbers k_1, k_2. Hence,

$$\begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix} = \begin{pmatrix} k_1e^{ct} \\ k_2e^{dt} \end{pmatrix} = k_1e^{ct}\begin{pmatrix} 1 \\ 0 \end{pmatrix} + k_2e^{dt}\begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

This implies that the general eigenfunction solution to $\mathbf{x}' = A\mathbf{x}$ is

$$\mathbf{x}(t) = c_1e^{\lambda_1 t}\mathbf{v}_1 + \cdots + c_ne^{\lambda_n t}\mathbf{v}_n$$

given that $\mathbf{v}_1, \ldots, \mathbf{v}_n$ are nonzero. Now, $\mathbf{v}$ and λ *must be an eigenvector-eigenvalue pair* for the solution to be true.

$$\begin{aligned} \mathbf{x}'(t) = A\mathbf{x}(t) = A\mathbf{v}e^{\lambda t} &= \left(\mathbf{v}e^{\lambda t}\right)' \\ \Rightarrow A\mathbf{v}e^{\lambda t} &= \lambda\mathbf{v}e^{\lambda t} \\ \Rightarrow A\mathbf{v} &= \lambda\mathbf{v} \end{aligned}$$

> ⓘ Note
>
> To solve for $c_1, \ldots, c_n$, we need an **initial condition**. That is, some vector $\mathbf{x}_0$ such that $\mathbf{x}(0) = \mathbf{x}_0$.

Example 5.6.1

What is the general solution to the following system of differential equations?

$$\frac{\mathrm{d}x}{\mathrm{d}t} = -x + 2y$$
$$\frac{\mathrm{d}y}{\mathrm{d}t} = 2x + 2y$$

$$\begin{pmatrix} x \\ y \end{pmatrix}' = \begin{pmatrix} -1 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

$$\begin{vmatrix} -1-\lambda & 2 \\ 2 & 2-\lambda \end{vmatrix} = 0$$

$$\begin{aligned} (-1-\lambda)(2-\lambda) - (2)(2) &= 0 \\ \lambda^2 - \lambda - 2 - 4 &= 0 \\ \lambda^2 - \lambda - 6 &= 0 \\ (\lambda+2)(\lambda-3) &= 9 \end{aligned}$$

$$\lambda_1 = -2$$

$$\begin{aligned} \big(-1-(-2)\big)x + 2y &= 0 \\ 2x + \big(2-(-2)\big)y &= 0 \end{aligned}$$

$$\begin{aligned} x + 2y &= 0 \\ 2x + 4y &= 0 \end{aligned}$$

$$\therefore \mathbf{v}_1 = \begin{pmatrix} -2 \\ 1 \end{pmatrix}$$

$$\lambda_2 = 3$$

$$\begin{aligned} (-1-3)x + 2y &= 0 \\ 2x + (2-3)y &= 0 \end{aligned}$$

$$\begin{aligned} -4x + 2y &= 0 \\ 2x - y &= 0 \end{aligned}$$

$$\therefore \mathbf{v}_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = c_1 e^{-2t} \begin{pmatrix} -2 \\ 1 \end{pmatrix} + c_2 e^{3t} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

◇

Phase Planes

We use phase planes to study all the trajectories of solutions to systems of differential equations in $\mathbb{R}^2$ for $t \geq 0$.

- The trajectories of the eigenfunctions lie in the eigenspaces of A. Other trajectories will approach the dominant eigenspace asymptotically.
- An eigenfunction will decay to the origin as $t \to \infty$ **if its associated eigenvalue is negative**.
- An eigenfunction will grow away from the origin as $t \to \infty$ **if its associated eigenvalue is positive**.
- Other trajectories will decay or grow depending on the eigenfunction they approach.

The origin has different names depending on whether the eigenfunctions approach it.

- The origin is a **saddle point** when A has positive and negative eigenvalues.
- The origin is a **sink** or **attractor** when A has negative eigenvalues.
  - Other trajectories will approach the eigenfunction associated with the greater eigenvalue.
- The origin is a **source** or **repeller** when A has positive eigenvalues.
  - Other trajectories will approach the eigenfunction associated with the least eigenvalue.

Complex Solutions

As established in previous chapters, complex eigenvalues have actual interpretations. In this case, complex solutions lead to *spiraling eigenfunctions and trajectories*. Using **Euler's formula**, we can graph these complex trajectories.

Definition 5.6.2 (Euler's Formula)

$$e^{ix} = \cos(x) + i\sin(x)$$

♣

- While this formula doesn't eliminate the i, we can eventually cancel them out due to the following:
  - Eigenspaces are closed under addition and scalar multiplication.
  - i is a scalar.

5.7 Applications to Markov Chains

Definition 5.7.1

A **probability vector** has nonnegative entries that sum to 1. A **stochastic matrix** is a square matrix whose columns are probability vectors.

- The transition matrices we covered in (1.10) *were* stochastic matrices.

Theorem 5.7.2

A stochastic matrix must have an eigenvalue equal to 1.

Proof. Let P be an $n \times n$ stochastic matrix. Thus, each row of $P^{\top}$ has entries that sum to 1. Furthermore, each row of $P^{\top} - 1I$ has entries that sum to 0, or equivalently, the sum of every column of $P^{\top} - 1I$ is $\mathbf{0}$. Because the columns of $P^{\top} - 1I$ are linearly dependent, $\det(P - 1I) = 0$ by the Invertible Matrix Theorem. By definition, 1 is an eigenvalue for $P^{\top}$, and by extension, and eigenvalue for P. □

- This makes sense, as we studied steady-state vectors in (1.10), which were essentially eigenvectors associated with an eigenvalue of 1.

6 Orthogonality and Least Squares

6.1 Inner Product, Length, and Orthogonality

Inner Product

Definition 6.1.1

Given vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, the **inner product** or dot product of $\mathbf{u}$ and $\mathbf{v}$ is defined as follows:

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^{\mathsf{T}}\mathbf{v}$$

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=0}^{n} u_i v_i$$

♣

Theorem 6.1.2

Let $\mathbf{u}, \mathbf{v}, \mathbf{w}$ be vectors in $\mathbb{R}^n$ and let c be some scalar.

1) $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
2) $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$
3) $(c\mathbf{u}) \cdot \mathbf{v} = c(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u} \cdot (c\mathbf{v})$
4) $\mathbf{u} \cdot \mathbf{u} \geq 0$
5) $\mathbf{u} \cdot \mathbf{u} = 0 \Leftrightarrow \mathbf{u} = \mathbf{0}$

♡

Magnitude and Distance

Definition 6.1.3

The **length**, **magnitude**, or **norm** of some vector $\mathbf{v} \in \mathbb{R}^n$ is a nonnegative scalar given by

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$$

♣

- This also implies that $\|\mathbf{v}\|^2 = \mathbf{v} \cdot \mathbf{v}$.
- A **unit vector** is a vector with a magnitude of 1. We can **normalize** a vector to convert it to a unit vector:

$$\hat{v} = \frac{\mathbf{v}}{\|\mathbf{v}\|}$$

Definition 6.1.4

Given two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$, the **distance between u and v**, denoted $\text{dist}(\mathbf{u}, \mathbf{v})$, is defined as follows:

$$\text{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

♣

- This definition is analogous to the Euclidean distance between points in $\mathbb{R}^2$ and $\mathbb{R}^3$.

Orthogonality

Like with distance in the previous chapter, orthogonality in $\mathbb{R}^n$ is an analog for orthogonality in $\mathbb{R}^2$ and $\mathbb{R}^3$.

Proposition 6.1.5

Two vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ are **orthogonal** or **perpendicular** to each other if $\mathbf{u} \cdot \mathbf{v} = 0$.

♠

Proof. Let $\mathbf{u}$ and $\mathbf{v}$ be two mutually orthogonal nonzero vectors in $\mathbb{R}^n$. By the law of cosines:

$$\begin{aligned}
\|\mathbf{u} - \mathbf{v}\|^2 &= \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\|\cos(\theta) \\
\|\mathbf{u} - \mathbf{v}\|^2 - \|\mathbf{u}\|^2 - \|\mathbf{v}\|^2 &= -2\|\mathbf{u}\|\|\mathbf{v}\|\cos(\theta) \\
\|\mathbf{u}\|\|\mathbf{v}\|\cos(\theta) &= \frac{\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - \|\mathbf{u} - \mathbf{v}\|^2}{2} \\
\|\mathbf{u}\|\|\mathbf{v}\|\cos(\theta) &= \frac{(\mathbf{u} \cdot \mathbf{u}) + (\mathbf{v} \cdot \mathbf{v}) - ((\mathbf{u} - \mathbf{v}) \cdot (\mathbf{u} - \mathbf{v}))}{2} \\
\|\mathbf{u}\|\|\mathbf{v}\|\cos(\theta) &= \frac{(\mathbf{u} \cdot \mathbf{u}) + (\mathbf{v} \cdot \mathbf{v}) - (\mathbf{u} \cdot \mathbf{u} - \mathbf{u} \cdot \mathbf{v} - \mathbf{v} \cdot \mathbf{u} + \mathbf{v} \cdot \mathbf{v})}{2} \\
\|\mathbf{u}\|\|\mathbf{v}\|\cos(\theta) &= \frac{\mathbf{u} \cdot \mathbf{v} + \mathbf{v} \cdot \mathbf{u}}{2} \\
\|\mathbf{u}\|\|\mathbf{v}\|\cos(\theta) &= \frac{\mathbf{u} \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}}{2} \\
\|\mathbf{u}\|\|\mathbf{v}\|\cos(\theta) &= \mathbf{u} \cdot \mathbf{v} \\
\|\mathbf{u}\|\|\mathbf{v}\|\cos\left(\frac{\pi}{2}\right) &= \mathbf{u} \cdot \mathbf{v} \\
\mathbf{u} \cdot \mathbf{v} &= 0
\end{aligned}$$

□

- A vector may be orthogonal to a subspace of $\mathbb{R}^n$ if its orthogonal to all vectors in the subspace.

Definition 6.1.6

Given subspace W of $\mathbb{R}^n$, the **orthogonal complement** of W, denoted $W^\perp$, is the set of all vectors that are orthogonal to W.

♣

Say we create a matrix A such that its columns make a basis for W, a subspace of $\mathbb{R}^n$. Additionally, let $\mathbf{x} \in \mathbb{R}^n$ and be orthogonal to W.

- $W = \text{Col } A$
- $W^\perp = \text{Nul } A^\mathsf{T}$
- $(\text{Col } A)^\perp = \text{Nul } A^\mathsf{T}$
- $\mathbf{x} \in \text{Nul } A^\mathsf{T}$
- $\mathbf{x} \in (\text{Row } A)^\perp$

Thus, we can create $W^\perp$ by solving $A^\mathsf{T}\mathbf{x} = \mathbf{0}$.

Example 6.1.7

If $W = \text{Span}\left\{\begin{pmatrix}1\\1\\1\end{pmatrix}, \begin{pmatrix}1\\2\\3\end{pmatrix}\right\}$, find $W^\perp$.

$$A = \begin{pmatrix}1 & 1\\1 & 2\\1 & 3\end{pmatrix}$$

$$A^\mathsf{T}\mathbf{x} = \mathbf{0}$$

$$\left(\begin{array}{ccc|c}1 & 1 & 1 & 0\\1 & 2 & 3 & 0\end{array}\right) \xrightarrow[R_2 - R_1 \to R_2]{} \left(\begin{array}{ccc|c}1 & 1 & 1 & 0\\0 & 1 & 2 & 0\end{array}\right) \xrightarrow[R_1 - R_2 \to R_1]{} \left(\begin{array}{ccc|c}1 & 0 & -1 & 0\\0 & 1 & 2 & 0\end{array}\right)$$

$$\therefore W^\perp = \text{Span}\left\{\begin{pmatrix}1\\-2\\1\end{pmatrix}\right\}$$

◇

6.2 Orthogonal Sets

Definition 6.2.1

An **orthogonal set** is a set of vectors $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_p\} \in \mathbb{R}^n$ such that $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ for $i \neq j$.

Theorem 6.2.2

If $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$, then S is linearly independent, and by extension, a basis for the subspace spanned by S.

Proof. Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ be an orthogonal set of nonzero vectors. Now, suppose

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_p\mathbf{u}_p = 0$$

where $c_1, c_2, \ldots, c_p$ are any scalars. For each i^{th} vector in S,

$$\begin{aligned} \left(c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_i\mathbf{u}_i + \cdots + c_p\mathbf{u}_p\right) \cdot \mathbf{u}_i &= 0 \\ c_1\left(\mathbf{u}_1 \cdot \mathbf{u}_i\right) + c_2\left(\mathbf{u}_2 \cdot \mathbf{u}_i\right) + \cdots + c_i\left(\mathbf{u}_i \cdot \mathbf{u}_i\right) + \cdots + c_p\left(\mathbf{u}_p \cdot \mathbf{u}_i\right) &= 0 \\ c_i\left\|\mathbf{u}_i\right\|^2 &= 0 \\ c_i &= 0 \end{aligned}$$

□

Definition 6.2.3

An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is an orthogonal set that forms a basis for W.

Theorem 6.2.4

Let $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $\mathbf{y} \in W$,

$$\mathbf{y} = c_1\mathbf{u}_1 + \cdots + c_i\mathbf{u}_i + \cdots + c_p\mathbf{u}_p$$

where

$$c_i = \frac{\mathbf{y} \cdot \mathbf{u}_i}{\mathbf{u}_i \cdot \mathbf{u}_i}$$

Proof. Let $B = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each i^{th} vector in B,

$$\begin{aligned}
\mathbf{y} &= c_1\mathbf{u}_1 + \cdots + c_i\mathbf{u}_i + \cdots + c_p\mathbf{u}_p \\
\mathbf{y} \cdot \mathbf{u}_i &= \left(c_1\mathbf{u}_1 + \cdots + c_i\mathbf{u}_i + \cdots + c_p\mathbf{u}_p\right) \cdot \mathbf{u}_i \\
\mathbf{y} \cdot \mathbf{u}_i &= c_1\left(\mathbf{u}_1 \cdot \mathbf{u}_i\right) + c_2\left(\mathbf{u}_2 \cdot \mathbf{u}_i\right) + \cdots + c_i\left(\mathbf{u}_i \cdot \mathbf{u}_i\right) + \cdots + c_p\left(\mathbf{u}_p \cdot \mathbf{u}_i\right) \\
\mathbf{y} \cdot \mathbf{u}_i &= c_i\left(\mathbf{u}_i \cdot \mathbf{u}_i\right) \\
c_i &= \frac{\mathbf{y} \cdot \mathbf{u}_i}{\mathbf{u}_i \cdot \mathbf{u}_i}
\end{aligned}$$

□

Example 6.2.5

Find an orthogonal basis for $\mathbb{R}^2$ that uses $\begin{pmatrix} 1 \\ 3 \end{pmatrix}$. Then, find the coordinates of $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ relative to the orthogonal basis.

$$\begin{aligned}
\mathbf{u} &= \begin{pmatrix} 1 \\ 3 \end{pmatrix} \\
\mathbf{u} \cdot \mathbf{v} &= 0 \\
v_1 + 3v_2 &= 0 \\
v_1 &= -3v_2 \\
\mathbf{v} &= \begin{pmatrix} -3 \\ 1 \end{pmatrix} \\
B &= \left\{ \begin{pmatrix} 1 \\ 3 \end{pmatrix}, \begin{pmatrix} -3 \\ 1 \end{pmatrix} \right\}
\end{aligned}$$

$$c_1 = \frac{\begin{pmatrix} 3 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix}}{\begin{pmatrix} 1 \\ 3 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 3 \end{pmatrix}} = \frac{15}{10} = \frac{3}{2} \qquad c_2 = \frac{\begin{pmatrix} 3 \\ 4 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 1 \end{pmatrix}}{\begin{pmatrix} -3 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} -3 \\ 1 \end{pmatrix}} = -\frac{5}{10} = -\frac{1}{2}$$

$$\begin{pmatrix} 3 \\ 4 \end{pmatrix}_B = \begin{pmatrix} \frac{3}{2} \\ -\frac{1}{2} \end{pmatrix}$$

Orthogonal Projection

Say we want to decompose the vector some vector $\mathbf{y} \in \mathbb{R}^n$ into the sum of two vectors: a multiple of $\mathbf{u}$ (some vector in $\mathbb{R}^n$) and a vector orthogonal to $\mathbf{u}$. Then,

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$$

where $\hat{\mathbf{y}} = \alpha\mathbf{u}$ for some scalar α and $\mathbf{z}$ is orthogonal to $\mathbf{u}$.

$$\begin{aligned}
\mathbf{y} \cdot \mathbf{u} &= (\hat{\mathbf{y}} + \mathbf{z}) \cdot \mathbf{u} \\
\mathbf{y} \cdot \mathbf{u} &= \alpha\mathbf{u} \cdot \mathbf{u} + \mathbf{z} \cdot \mathbf{u} \\
\mathbf{y} \cdot \mathbf{u} &= \alpha(\mathbf{u} \cdot \mathbf{u}) \\
\alpha &= \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}
\end{aligned}$$

Thus, this decomposition only holds for $\hat{\mathbf{y}} = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}$. This vector is known as the **orthogonal projection** of $\mathbf{y}$ onto $\mathbf{u}$. If the subspace L is the line spanned by $\mathbf{u}$, $\hat{\mathbf{y}}$ is also the **orthogonal projection** of $\mathbf{y}$ onto L, denoted $\operatorname{proj}_L \mathbf{y}$. Meanwhile, the vector $\mathbf{z}$ is the component of $\mathbf{y}$ orthogonal to $\mathbf{u}$.

- $\|\mathbf{z}\|$ is the **distance of $\mathbf{y}$ from** L.

Orthonormal Sets

Definition 6.2.6

An **orthonormal set** is a set of orthogonal unit vectors $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\} \in \mathbb{R}^n$. By theorem (6.2.2), it is also an **orthonormal basis** for the subspace it spans.

- Because all the vectors in an orthonormal set are unit vectors, orthogonal projections are simplified because for any vector $\mathbf{u}$ in the set, $\mathbf{u} \cdot \mathbf{u} = 1$.

Theorem 6.2.7

An $m \times n$ matrix U has orthonormal columns $\Leftrightarrow U^\top U = I_n$.

Proof. Let U be an $m \times n$ matrix with orthonormal columns $\mathbf{u}_1, \dots, \mathbf{u}_n$.

$$U^\top U = \begin{pmatrix} \mathbf{u}_1^\top \\ \mathbf{u}_2^\top \\ \vdots \\ \mathbf{u}_n^\top \end{pmatrix} \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_n \end{pmatrix}$$

$$U^\top U = \begin{pmatrix} \mathbf{u}_1^\top\mathbf{u}_1 & \mathbf{u}_1^\top\mathbf{u}_2 & \cdots & \mathbf{u}_1^\top\mathbf{u}_n \\ \mathbf{u}_2^\top\mathbf{u}_1 & \mathbf{u}_2^\top\mathbf{u}_2 & \cdots & \mathbf{u}_2^\top\mathbf{u}_n \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{u}_n^\top\mathbf{u}_1 & \mathbf{u}_n^\top\mathbf{u}_2 & \cdots & \mathbf{u}_n^\top\mathbf{u}_n \end{pmatrix} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix} = I_n \qquad \square$$

Theorem 6.2.8

Let U be an $m \times n$ matrix with orthonormal columns, and let $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. Then

1) $\|U\mathbf{x}\| = \|\mathbf{x}\|$
2) $(U\mathbf{x}) \cdot (U\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$
3) $(U\mathbf{x}) \cdot (U\mathbf{y}) = 0 \Leftrightarrow \mathbf{x} \cdot \mathbf{y} = 0$

- Properties (1) and (3) indicate that the linear mapping $\mathbf{x} \mapsto U\mathbf{x}$ preserves magnitude and orthogonality. The rotational transformation from (1.9) follows those properties, and the associated standard matrix is indeed orthonormal by the Pythagorean identity.

Definition 6.2.9

An **orthogonal matrix** is an $n \times n$ matrix U with orthonormal columns such that

$$U^{-1} = U^{\top}$$

6.3 Orthogonal Projections

In (6.2), we covered orthogonal projections onto vectors. Now, we will generalize that definition to subspaces of $\mathbb{R}^n$.

Theorem 6.3.1 (Orthogonal Decomposition Theorem)

Let W be a subspace of $\mathbb{R}^n$. Then, each $\mathbf{y} \in \mathbb{R}^n$ can be written uniquely as

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$$

where $\hat{\mathbf{y}} \in W$ and $\mathbf{z} \in W^\perp$. Now, if $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ is an orthogonal basis for W:

$$\hat{\mathbf{y}} = \operatorname{proj}_W \mathbf{y} = \frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\mathbf{u}_1 + \cdots + \frac{\mathbf{y} \cdot \mathbf{u}_p}{\mathbf{u}_p \cdot \mathbf{u}_p}\mathbf{u}_p$$

and $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

Properties of Orthogonal Projections

Theorem 6.3.2 (The Best Approximation Theorem)

Let W be a subspace of $\mathbb{R}^n$, $\mathbf{y} \in \mathbb{R}^n$, and $\hat{\mathbf{y}} = \operatorname{proj}_W \mathbf{y}$. Then, $\hat{\mathbf{y}}$ is the *closest point* in W to $\mathbf{y}$, meaning

$$\|\mathbf{y} - \hat{\mathbf{y}}\| < \|\mathbf{y} - \mathbf{v}\| \quad \forall \mathbf{v} \neq \hat{\mathbf{y}} \in W$$

- If $\mathbf{y} \in W$, then $\hat{\mathbf{y}} = \operatorname{proj}_W \mathbf{y} = \mathbf{y}$.

Example 6.3.3

Find the closest point to $\begin{pmatrix}1\\2\\3\end{pmatrix}$ on the plane W spanned by $\left\{\begin{pmatrix}1\\0\\1\end{pmatrix}, \begin{pmatrix}1\\2\\-1\end{pmatrix}\right\}$.

$$\hat{\mathbf{y}} = \frac{\begin{pmatrix}1\\2\\3\end{pmatrix} \cdot \begin{pmatrix}1\\0\\1\end{pmatrix}}{\begin{pmatrix}1\\0\\1\end{pmatrix} \cdot \begin{pmatrix}1\\0\\1\end{pmatrix}}\begin{pmatrix}1\\0\\1\end{pmatrix} + \frac{\begin{pmatrix}1\\2\\3\end{pmatrix} \cdot \begin{pmatrix}1\\2\\-1\end{pmatrix}}{\begin{pmatrix}1\\2\\-1\end{pmatrix} \cdot \begin{pmatrix}1\\2\\-1\end{pmatrix}}\begin{pmatrix}1\\2\\-1\end{pmatrix}$$

$$\hat{\mathbf{y}} = \frac{4}{2}\begin{pmatrix}1\\0\\1\end{pmatrix} + \frac{2}{6}\begin{pmatrix}1\\2\\-1\end{pmatrix} = \begin{pmatrix}2\\0\\2\end{pmatrix} + \begin{pmatrix}1/3\\2/3\\-1/3\end{pmatrix}$$

$$\hat{\mathbf{y}} = \begin{pmatrix}7/3\\2/3\\5/3\end{pmatrix}$$

Theorem 6.3.4

Let W be a subspace of $\mathbb{R}^n$ and let $\mathbf{y} \in W$. If $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ is an orthonormal basis for W:

$$\operatorname{proj}_W \mathbf{y} = (\mathbf{y} \cdot \mathbf{u}_1)\mathbf{u}_1 + \cdots + (\mathbf{y} \cdot \mathbf{u}_p)\mathbf{u}_p$$

Let U be an orthogonal matrix such that $U = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_p \end{pmatrix}$. Then,

$$\operatorname{proj}_W \mathbf{y} = UU^\top \mathbf{y}$$

6.4 The Gram-Schmidt Process

The **Gram-Schmidt Process** is an algorithm for creating orthogonal bases for a subspace of $\mathbb{R}^n$. It hinges on orthogonal decompositions by continuously creating vectors that are orthogonal to the previous ones.

Algorithm 6.4.1 (The Gram-Schmidt Process)

Given a basis $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_p\}$ for a nonzero subspace W of $\mathbb{R}^n$:

$$\begin{aligned}
\mathbf{v}_1 &= \mathbf{x}_1 \\
\mathbf{v}_2 &= \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 \\
\mathbf{v}_3 &= \mathbf{x}_3 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_3 \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_1}\mathbf{v}_2 \\
&\vdots \\
\mathbf{v}_p &= \mathbf{x}_p - \frac{\mathbf{x}_p \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1}\mathbf{v}_1 - \frac{\mathbf{x}_p \cdot \mathbf{v}_2}{\mathbf{v}_2 \cdot \mathbf{v}_1}\mathbf{v}_2 - \cdots - \frac{\mathbf{x}_p \cdot \mathbf{v}_{p-1}}{\mathbf{v}_{p-1} \cdot \mathbf{v}_{p-1}}\mathbf{v}_{p-1}
\end{aligned}$$

The resulting set $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_p\}$ is an orthogonal basis for W. Additionally, by definition,

$$\operatorname{Span}\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k\} = \operatorname{Span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k\} \quad \forall 1 \leq k \leq p$$

6.5 Least-Squares Problems

In an inconsistent system, we may be interested in some solution that gets closest to the desired answer. In other words, finding some $\mathbf{x}$ that minimizes $\|\mathbf{b} - A\mathbf{x}\|$. This is known as a **general least-squares problem**.

Remark

The term *least-squares* refers to $\|\mathbf{b} - A\mathbf{x}\|$ being the square root of a sum of squares.

Definition 6.5.1

Given an $m \times n$ matrix A, a **least-squares solution** to $A\mathbf{x} = \mathbf{b}$ is some $\hat{\mathbf{x}} \in \mathbb{R}^n$ such that

$$\|\mathbf{b} - A\hat{\mathbf{x}}\| \leq \|\mathbf{b} - A\mathbf{x}\| \quad \forall x \in \mathbb{R}^n$$

♣

- So, if $\hat{\mathbf{x}}$ refers to the best approximation, then the following should be true:

$$A\hat{\mathbf{x}} = \text{proj}_{\text{Col } A}\, \mathbf{b} = \hat{\mathbf{b}}$$

- It follows that $\mathbf{b} - \hat{\mathbf{b}}$ is orthogonal to Col A. Therefore,

$$\begin{aligned} \mathbf{b} - \hat{\mathbf{b}} &\in (\text{Col } A)^{\perp} \\ \mathbf{b} - \hat{\mathbf{b}} &\in \text{Nul } A^{\mathsf{T}} \\ A^{\mathsf{T}}\left(\mathbf{b} - \hat{\mathbf{b}}\right) &= \mathbf{0} \\ A^{\mathsf{T}}\mathbf{b} - A^{\mathsf{T}}\hat{\mathbf{b}} &= \mathbf{0} \\ A^{\mathsf{T}}\hat{\mathbf{b}} &= A^{\mathsf{T}}\mathbf{b} \\ A^{\mathsf{T}}A\hat{\mathbf{x}} &= A^{\mathsf{T}}\mathbf{b} \end{aligned}$$

The distance $\left\|\mathbf{b} - \hat{\mathbf{b}}\right\|$ is known as the **least-squares error**.

Example 6.5.2

Create a linear regression for the points $(0, 0), (1, 1), (3, 1)$.

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} m \\ k \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} m \\ k \end{pmatrix} = \begin{pmatrix} 0 & 1 & 2 \\ 1 & 1 & 1 \end{pmatrix}\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$$

$$\left(\begin{array}{cc|c} 5 & 3 & 3 \\ 3 & 3 & 2 \end{array}\right) \xrightarrow[\text{rref}]{} \left(\begin{array}{cc|c} 1 & 0 & \frac{1}{2} \\ 0 & 1 & \frac{1}{6} \end{array}\right) \Rightarrow \quad y = \frac{1}{2}x + \frac{1}{6}$$

◊

7 Symmetric Matrices and Quadratic Forms

7.1 Diagonalization of Symmetric Matrices

A **symmetric matrix** is some matrix A such that $A = A^{\top}$.

Theorem 7.1.1

If A is symmetric, then any two eigenvectors from different eigenspaces will be orthogonal.

Proof. Let $\mathbf{v}_1$ and $\mathbf{v}_2$ be eigenvectors that correspond to different eigenvalues λ_1 and λ_2 of some symmetric matrix A, respectively.

$$\begin{aligned}
\lambda_1 \mathbf{v}_1 \cdot \mathbf{v}_2 &= (\lambda_1 \mathbf{v}_1)^{\top} \mathbf{v}_2 \\
\lambda_1 \mathbf{v}_1 \cdot \mathbf{v}_2 &= (A \mathbf{v}_1)^{\top} \mathbf{v}_2 \\
\lambda_1 \mathbf{v}_1 \cdot \mathbf{v}_2 &= \mathbf{v}_1^{\top} A^{\top} \mathbf{v}_2 \\
\lambda_1 \mathbf{v}_1 \cdot \mathbf{v}_2 &= \mathbf{v}_1^{\top} (A \mathbf{v}_2) \\
\lambda_1 \mathbf{v}_1 \cdot \mathbf{v}_2 &= \mathbf{v}_1^{\top} (\lambda_2 \mathbf{v}_2) \\
\lambda_1 \mathbf{v}_1 \cdot \mathbf{v}_2 &= \lambda_2 \mathbf{v}_1 \cdot \mathbf{v}_2 \\
(\lambda_1 - \lambda_2) \mathbf{v}_1 \cdot \mathbf{v}_2 &= 0 \\
\mathbf{v}_1 \cdot \mathbf{v}_2 &= 0 \quad \text{Because } \lambda_1 - \lambda_2 \neq 0
\end{aligned}$$

□

An $n \times n$ matrix A is **orthogonally diagonalizable** if there exists an orthogonal matrix P and a diagonal matrix D such that

$$A = PDP^{-1} = PDP^{\top}$$

Theorem 7.1.2

An $n \times n$ matrix A is orthogonally diagonalizable $\Leftrightarrow$ A is symmetric.

Proof. Let A be an orthogonally diagonalizable matrix.

$$\begin{aligned}
A &= PDP^{\top} \\
A^{\top} &= (PDP^{\top})^{\top} \\
A^{\top} &= (P^{\top})^{\top} D^{\top} P^{\top} \\
A^{\top} &= PDP^{\top} \\
A^{\top} &= A
\end{aligned}$$

□

Important

Recall that an orthogonal matrix has *orthonormal columns*, so eigenvectors should be normalized before forming P when orthogonally diagonalizing a matrix.

The Spectral Theorem

Theorem 7.1.3 (The Spectral Theorem for Symmetric Matrices)

An $n \times n$ symmetric matrix A has the following properties:
1) A has n real eigenvalues, including multiplicities.
2) The dimension of each eigenvalue's eigenspace equals its multiplicity.
3) The eigenspaces are mutually orthogonal.
4) A is orthogonally diagonalizable.

Furthermore, the **spectral decomposition** of A is as follows:

$$A = \lambda_1 \mathbf{u}_1 \mathbf{u}_1^\top + \lambda_2 \mathbf{u}_2 \mathbf{u}_2^\top + \cdots + \lambda_1 \mathbf{u}_n \mathbf{u}_n^\top$$

Remark

The *spectrum* of a matrix refers to the set of its eigenvalues.

Say the spectral decomposition of A is ordered in descending order such that $|\lambda_1| \geq |\lambda_2| \geq \cdots \geq |\lambda_n|$. We can *approximate* A using $\lambda_1, \lambda_2, \ldots$, becoming more accurate (and eventually exact) as we use more eigenvalues.

7.2 Quadratic Forms

A **quadratic form** is a function Q defined on $\mathbb{R}^n$ such that every term has a degree of 2. It can be expressed in the form $Q(\mathbf{x}) = \mathbf{x}^\top A\mathbf{x}$ where A is a symmetric matrix.

- For instance, the quadratic form associated with symmetric matrix $A = \begin{pmatrix} 1 & 2 \\ 2 & -3 \end{pmatrix}$ would be:

$$\begin{pmatrix} x_1 & x_2 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 2 & -3 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_1^2 - 4x_1x_2 - 3x_2^2$$

- When given the quadratic form, the coefficients of the squared terms represent the main diagonal. Meanwhile, the coefficients of the cross-product terms can be split into two terms, representing a pair of entries on opposite sides of the main diagonal.
- Given $Q(x) = x_1^2 + 4x_1x_2 + 3x_1x_3 - x_3^2$,

$$\begin{aligned}
Q(x) &= x_1^2 + 4x_1x_2 + 3x_1x_3 - x_3^2 \\
Q(x) &= x_1^2 + \left(2x_1x_2 + 2x_2x_1\right) + \left(\frac{3}{2}x_1x_3 + \frac{3}{2}x_3x_1\right) - x_3^2 \\
\therefore A &= \begin{pmatrix} 1 & 2 & 3/2 \\ 2 & 0 & 0 \\ 3/2 & 0 & -1 \end{pmatrix}
\end{aligned}$$

Change of Variable

If we want to eliminate cross-product terms in quadratic forms, we can implement a **change of variable** of the following form

$$\mathbf{x} = P\mathbf{y} \quad \text{or} \quad \mathbf{y} = P^{-1}\mathbf{x}$$

where P is invertible. Substituting this into the general quadratic form, we get:

$$\begin{aligned}
\mathbf{x}^\top A\mathbf{x} &= (P(\mathbf{y}))^\top A(P\mathbf{y}) \\
\mathbf{x}^\top A\mathbf{x} &= \left(\mathbf{y}^\top P^\top\right) A(P\mathbf{y}) \\
\mathbf{x}^\top A\mathbf{x} &= \mathbf{y}^\top\left(P^\top AP\right)\mathbf{y} \\
\mathbf{x}^\top A\mathbf{x} &= \mathbf{y}^\top D\mathbf{y} \quad \text{Because } A \text{ is orthogonally diagonalizable}
\end{aligned}$$

- The columns of P are known as the **principle axes** of the quadratic form. Essentially the coordinate vector of $\mathbf{x}$ relative to the orthonormal basis formed by the columns P is $\mathbf{y}$.

Classifying Quadratic Forms

Quadratic forms have different classifications depending on the sign of their output.

Definition 7.2.1

A quadratic form Q is
1) **Positive definite** if $Q(\mathbf{x}) > 0$ for all $\mathbf{x} \neq \mathbf{0}$.
2) **Positive semidefinite** if $Q(\mathbf{x}) \geq 0$ for all $\mathbf{x} \neq \mathbf{0}$.
3) **Negative definite** if $Q(\mathbf{x}) < 0$ for all $\mathbf{x} \neq \mathbf{0}$.
4) **Negative semidefinite** if $Q(\mathbf{x}) > 0$ for all $\mathbf{x} \neq \mathbf{0}$.
5) **Indefinite** if $Q(\mathbf{x})$ can be both positive and negative.

♣

Theorem 7.2.2

Let A be an $n \times n$ symmetric matrix. Then, the associated quadratic form $\mathbf{x}^{\mathsf{T}} A \mathbf{x}$ is
1) **Positive definite** $\Leftrightarrow$ the eigenvalues of A are all positive.
2) **Positive semidefinite** $\Leftrightarrow$ the eigenvalues of A are all nonnegative.
3) **Negative definite** $\Leftrightarrow$ the eigenvalues of A are all negative.
4) **Negative semidefinite** $\Leftrightarrow$ the eigenvalues of A are all less than or equal to zero.
5) **Indefinite** $\Leftrightarrow$ A has both positive and negative eigenvalues.

♡

This is why the change of variable also shows classifications of quadratic forms.

7.3 Constrained Optimization

In applications, finding the maximum and minimum values of a quadratic form $Q(\mathbf{x})$ under a **constraint** can be necessary. Typically, the constraint is tha $\mathbf{x}$ must be a unit vector. This constraint has many equivalent forms:

$$\|\mathbf{x}\| = 1 \qquad \|\mathbf{x}\|^2 = 1 \qquad \mathbf{x}^\mathsf{T}\mathbf{x} = 1$$

$$x_1^2 + x_2^2 + \cdots + x_n^2 = 1$$

Without cross product-terms, maximum and minimum values are easy to calculate, as the largest coefficient corresponds to the maximum value and the smallest coefficient corresponds to the minimum value.

If the quadratic form has cross-product terms, then we can orthogonally diagonalize it to eliminate cross-product terms. However, because the eigenvalues correspond to the coefficients of the change of variable, *we can just find the eigenvalues if we are just interested in the extrema.*

Theorem 7.3.1

Let A be a symmetric matrix associated with the quadratic form $Q(\mathbf{x}) = \mathbf{x}^\mathsf{T} A\mathbf{x}$. Additionally, let the greatest eigenvalue be λ_{max} and the least eigenvalue be λ_{min} correspond to unit eigenvectors $\mathbf{u}_1$ an $\mathbf{u}_w$, respectively. Thus, the maximum value of Q is λ_{max} at $\mathbf{x} = \mathbf{u}_1$ and the minimum value of Q is λ_{min} at $\mathbf{x} = \mathbf{u}_1$.